

Test supermartingales are complete for analytic sequential hypotheses

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Abstract

E-processes provide a general framework for anytime-valid sequential testing, while test supermartingales form a more structured subclass whose validity can be verified through one-step conditional inequalities. In general, however, an e-process need not be a test supermartingale, and it is natural to ask when restricting attention to test supermartingales entails no loss of testing power.

We answer this question for a broad class of composite sequential hypotheses specified through admissible one-step conditional laws. After each finite history s , a correspondence $\mathcal{K}(s) \subseteq \Delta(\mathcal{Y})$ determines the distributions that may govern the next observation, with the local sets allowed to vary with the history and subject only to an analytic-graph regularity condition. The resulting null contains every path law generated by a universally measurable sequential selection from $\mathcal{K}$. We prove that every e-process valid for such an analytic sequential hypothesis is pointwise dominated by a Borel test supermartingale that is valid uniformly over the same null. Thus test supermartingales form a complete class for anytime-valid testing: arbitrary e-processes can be replaced, without loss of testing power, by test supermartingales.

Our result does not require local domination, convexity, or compactness of the admissible one-step sets, and includes examples not captured by previously studied finite-dimensional conditional-constraint models. We illustrate the scope of analytic sequential hypotheses via an array of examples.

1 Introduction

Anytime-valid inference allows evidence to be monitored continuously without fixing the stopping rule in advance (Howard et al., 2021; Ramdas et al., 2023; Grünwald et al., 2024). E-processes provide a general way to express this guarantee: their expected value remains bounded at every stopping time under every law in the null. This definition is flexible, but global—validity is a statement about all stopping rules under all null laws. Test supermartingales impose more structure, replacing this global requirement by a one-step conditional drift inequality at every history. That local form is useful both conceptually and operationally: the next move can be chosen from the data seen so far, and validity can be checked one step at a time. For example, in an online experiment comparing two treatments, an analyst may adapt the next test statistic or allocation rule after each new batch of outcomes. With a test supermartingale, it is enough to verify that the resulting next-step factor has conditional expectation at most one under the null; one does not have to re-establish optional-stopping validity over all possible future continuations after every adaptation. This local structure is one reason test supermartingales are especially convenient for constructing and optimizing sequential tests (Shafer et al., 2011; Ramdas et al.,

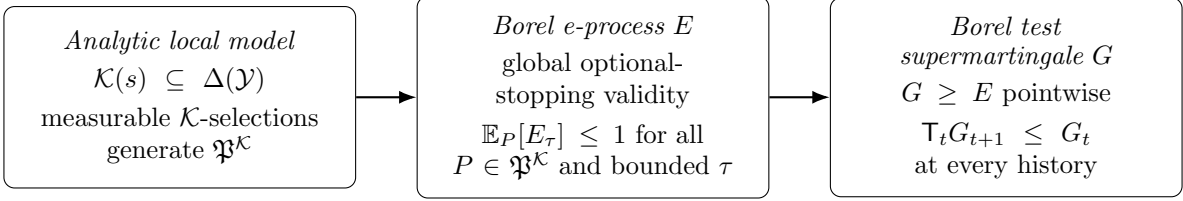


Figure 1: Architecture of the complete-class result. The analytic local model and universally measurable $\mathcal{K}$ -selections are formalized in Section 2.2 and Definition 2.1, and generate the sequential hypothesis $\mathfrak{P}^{\mathcal{K}}$ of Definition 2.3. Global optional-stopping validity is the e-process condition of Definition 2.4, equivalently (1). Theorem 5.1 replaces any such E by a Borel test supermartingale G (Definition 2.5) that pointwise dominates E in the sense of Definition 2.6 and satisfies the historywise upper-expectation inequality (7); the operator T_t is defined in (4). The construction proceeds through the **usa** Snell envelope of Theorem 3.4, the Borel reflection step of Proposition 4.2, and their interlacing in Proposition 4.3.

2023). Every test supermartingale is an e-process, but the converse need not hold (Ramdas et al., 2022). This motivates the question:

When can every valid e-process be replaced, without loss for testing, by a test supermartingale?

We answer this question for a broad class of sequential hypotheses described directly through their admissible one-step laws. After each finite history s , let $\mathcal{K}(s) \subseteq \Delta(\mathcal{Y})$ be the set of distributions allowed for the next observation. We assume that the graph of $\mathcal{K}$ is analytic and take the global null to contain every path law generated by a universally measurable sequential selection from these local sets. The formal local-to-global construction is given in Section 2.2, with $\mathcal{K}$ -selections defined in Definition 2.1 and the induced *analytic sequential hypothesis* in Definition 2.3. The local sets may vary with the history and need not be convex, compact, or dominated by a common reference measure. For analytic sequential hypotheses, we prove that the aforementioned question has an affirmative answer.

Theorem 1.1 (Informal statement of Theorem 5.1). *For every analytic sequential hypothesis, test supermartingales form a complete class for anytime-valid testing. Equivalently, every Borel e-process valid for the hypothesis is pointwise dominated, at no more than unit initial capital, by a Borel test supermartingale valid for the same hypothesis.*

The strength of the conclusion lies in the form of domination. Pointwise domination, formalized in Definition 2.6 and (3), is stronger than simultaneous almost-sure domination under the null: the dominating process is at least as large after every finite history, including histories that may have probability zero under every particular null law. Consequently, any threshold crossing of the original e-process is matched no later by the dominating test supermartingale. The formal statement is Theorem 5.1 in Section 5. Figure 1 summarizes the local-to-global structure of the result and points forward to the formal definitions and proof ingredients.

Obtaining this historywise conclusion while returning to the Borel class is the main technical issue. The proof has two distinct parts. A robust Snell recursion first converts optional-stopping validity into pointwise one-step supermartingale inequalities on the history space, of the form ultimately recorded in (7); the finite-horizon recursion is given in Protocol 1, and its infinite-horizon consequence is Theorem 3.4. Because optimization over an analytic graph preserves upper semianalyticity rather than Borel measurability, this first envelope need not itself be Borel. We then use a reflection argument to enlarge the process back into the Borel class

without increasing its stopping-time cost (Proposition 4.2). Reflection can destroy the one-step inequalities, so the two operations are alternated; Proposition 4.3 shows that their monotone limit is both Borel and a pointwise supermartingale. The same measurable-selection issue appears concretely in the latent-feasibility example of Section 6.6, where a continuous continuation payoff is sent to a non-Borel value function.

The same local formulation also accommodates familiar moment restrictions as well as genuinely distributional and analytic conditions. Bounded conditional means provide a familiar running case (Example 6.1) and already show that local domination can fail in a classical problem (Howard et al., 2021; Clerico, 2025). The lavender running-example boxes that appear along the way return to this same bounded-mean model as an illustration of the general definitions and constructions. Conditional first-order stochastic dominance (Section 6.3) and conditional sub-Gaussianity (Section 6.4) are statistically standard restrictions (Arnold et al., 2026; Howard et al., 2021); our examples show that the corresponding local classes are genuinely infinite-dimensional and lie beyond finite-feature conditional-constraint models. The conditional Wasserstein-ball example in Section 6.5 gives a distributional-stability model whose upper transition operator is a familiar distributionally robust optimization problem (Esfahani and Kuhn, 2018). The latent-feasibility model of Section 6.6 connects the analytic machinery to projection and measurable-selection phenomena that also arise in partial identification (Kline and Tamer, 2016; Beresteanu et al., 2012; Gu and Russell, 2023).

1.1 Related work

The role of nonnegative martingales and supermartingales as canonical objects for anytime-valid inference has been studied in several nearby settings. Ramdas et al. (2020) show, in particular, that under a point null every e-process is dominated by a nonnegative martingale, and develop a broader admissibility theory for composite hypotheses under domination conditions. A closer analogue of our question appears in Ramdas et al. (2022): under local domination and fork-convexity of the null family, a composite Snell-envelope construction shows that every safe e-process is dominated by a nonnegative supermartingale valid under the whole null. Our result addresses the same complete-class question under a different structural regime. We work from an analytic correspondence of admissible one-step laws and include all path laws generated by measurable sequential selections. This allows genuinely nondominated models and does not require the local sets $\mathcal{K}(s)$ to be convex. Neither framework contains the other in general.

A complementary line of work starts from one-step distributional constraints. Larsson et al. (2026) develop a broad complete-class theory for e-variables under measurable constraints, including moment restrictions and classes such as sub-Gaussian distributions. Passing from one step to a sequential null requires the constraints and the corresponding e-variables to be chosen predictably across histories. Clerico (2026) carries this out for finitely many conditional constraints and shows that every e-process is pointwise dominated by a predictable product of affine one-step e-variables. For conditional means, Clerico (2025) gives an especially explicit version of this phenomenon. Our local formulation overlaps with this perspective at the level of one-step constraints, since many such restrictions can be encoded directly through the admissible sets $\mathcal{K}(s)$. These results are complementary in emphasis: the finite-dimensional theory yields an explicit affine characterization when it applies, while we seek a complete-class result for local models that may not admit any finite-feature representation. The examples in Sections 6.3–6.6 include stochastic-order, sub-Gaussian, transport-based, and genuinely analytic restrictions.

Related complete-class phenomena also appear in other formulations of anytime-valid inference. For example, Vovk and Wang (2024) show that martingale-based procedures form a com-

plete class for merging sequential e-values. That problem takes sequential e-values themselves as the primitive objects, rather than a composite null over the observations, but it reinforces the recurring role of martingales as canonical representatives of anytime-valid procedures.

There is also a structural connection to rectangularity, pasting, and analytic product structure in robust decision theory and nondominated stochastic analysis. In finite event trees, the rectangularity of Epstein and Schneider (2003) has a local-product interpretation: the global family of priors contains all admissible branchwise recombinations of its one-step conditionals. Our formulation takes the local description as primitive and extends it to a general standard Borel observation space and infinite horizon, where branchwise choices must be made measurably. The analytic-graph assumption supplies the regularity needed for those selections. Thus our setup is best viewed as an analytically regular infinite-horizon local-product formulation, rather than as bare rectangularity itself.

A particularly close technical precedent is the analytic product structure used in nondominated mathematical finance and nonlinear expectations; see, e.g., Bouchard and Nutz (2015); Nutz and van Handel (2013). There too, global families of measures are built from local correspondences with analytic graphs and universally measurable selectors, usually for robust control, superhedging, or nonlinear dynamic programming. Here the same measurable structure is used to study whether an arbitrary anytime-valid process can be dominated by a single composite supermartingale with initial capital at most one.

2 Sequential model and testing framework

We work on a canonical discrete-time path space. The composite null is specified locally, through the set of conditional laws that may govern the next observation after each finite history. We then introduce e-processes and test supermartingales for the resulting family of path laws, and formalize the complete-class question studied in this paper. For background on e-values, e-processes, and anytime-valid testing, see, e.g., Ramdas et al. (2023) and Grünwald et al. (2024).

2.1 Canonical path space

Let $\mathcal{Y}$ be a Polish space, equipped with its Borel σ -field $\mathcal{B}(\mathcal{Y})$. For $t \geq 0$, let $\mathcal{Y}^t$ denote the space of histories of length t , with $\mathcal{Y}^0 := \{\emptyset\}$. The *finite-history space* is $\mathbf{H} := \bigsqcup_{t \geq 0} \mathcal{Y}^t$. Since $\mathcal{Y}$ is Polish, each $\mathcal{Y}^t$ is Polish, and with the natural disjoint-union topology $\mathbf{H}$ is again Polish. For $s \in \mathcal{Y}^t$ and $y \in \mathcal{Y}$, we write $s \oplus y \in \mathcal{Y}^{t+1}$ for concatenation.

The *path space* is $\Omega := \mathcal{Y}^{\mathbb{N}}$, equipped with the product Borel σ -field $\mathcal{F} := \mathcal{B}(\Omega)$. Let $Y_t(\omega) := \omega_t$ denote the coordinate process, and write $Y_{1:t} := (Y_1, \dots, Y_t)$. The *canonical filtration* is $\mathcal{F}_t := \sigma(Y_1, \dots, Y_t)$ for $t \geq 1$, with $\mathcal{F}_0 := \{\emptyset, \Omega\}$.

Let $\Delta(\mathcal{Y})$ denote the space of Borel probability measures on $\mathcal{Y}$, equipped with the topology of weak convergence and its Borel σ -field. For each $t \geq 0$, let $\mathcal{U}_t$ denote the *universal completion* of $\mathcal{B}(\mathcal{Y}^t)$, namely

$$\mathcal{U}_t := \bigcap_{Q \in \Delta(\mathcal{Y}^t)} \mathcal{B}(\mathcal{Y}^t)^Q,$$

where $\mathcal{B}(\mathcal{Y}^t)^Q$ denotes the Q -completion of $\mathcal{B}(\mathcal{Y}^t)$.

Throughout, a nonnegative process $X = (X_t)_{t \geq 0}$ is identified with its history-space representatives $X_t : \mathcal{Y}^t \rightarrow [0, \infty]$, so that its canonical realization is $X_t(Y_{1:t})$. We call X *Borel adapted* if each X_t is Borel measurable. Larger measurability classes will arise naturally in our proofs and will be introduced when needed.

2.2 Local models and the induced sequential family

We describe the null locally by specifying which one-step distributions are admissible after each finite history. Let $\mathcal{K} : \mathbf{H} \rightrightarrows \Delta(\mathcal{Y})$ be a nonempty-valued correspondence. For a history $s \in \mathbf{H}$, the set $\mathcal{K}(s)$ is interpreted as the collection of admissible laws for the next observation following s .

Our basic regularity assumption is that the graph

$$\text{Gr}(\mathcal{K}) := \{(s, Q) \in \mathbf{H} \times \Delta(\mathcal{Y}) : Q \in \mathcal{K}(s)\}$$

is *analytic*. We refer to such a correspondence as an *analytic local model*. Analytic-graph assumptions of this type are standard in universally measurable dynamic programming and in nondominated stochastic models; see, e.g., Bertsekas and Shreve (1996, Chapter 7), Nutz and van Handel (2013), and Bouchard and Nutz (2015). We do not assume that the sets $\mathcal{K}(s)$ are convex or compact, nor do we impose a common dominating measure.

A measurable choice of one admissible law at every history determines a probability law on the full path space.

Definition 2.1 ($\mathcal{K}$ -selection). *A $\mathcal{K}$ -selection is a sequence $\kappa = (\kappa_t)_{t \geq 0}$ such that, for every $t \geq 0$, the map $\kappa_t : \mathcal{Y}^t \rightarrow \Delta(\mathcal{Y})$ is $\mathcal{U}_t/\mathcal{B}(\Delta(\mathcal{Y}))$ -measurable and*

$$\kappa_t(s) \in \mathcal{K}(s) \quad \text{for every } s \in \mathcal{Y}^t.$$

For $A \in \mathcal{B}(\mathcal{Y})$, we write $\kappa_t(s, A) := \kappa_t(s)(A)$.

Since $\text{Gr}(\mathcal{K})$ is analytic and its sections are nonempty, the Jankov–von Neumann measurable selection theorem guarantees the existence of universally measurable selectors (see, e.g., Bertsekas and Shreve, 1996, Chapter 7). The use of universal rather than Borel measurability is important here: an analytic graph need not admit a Borel selector (see, e.g., Bertsekas and Shreve, 1996, Chapter 7).

Every $\mathcal{K}$ -selection induces a path law. Lemma 2.2 makes the passage from the local specification $\mathcal{K}$ to a global family of probability measures precise.

Lemma 2.2 (Path law induced by a $\mathcal{K}$ -selection). *For every $\mathcal{K}$ -selection $\kappa = (\kappa_t)_{t \geq 0}$, there exists a unique probability measure $P^\kappa \in \Delta(\Omega)$ such that, for every $n \geq 1$ and every bounded Borel function $f : \mathcal{Y}^n \rightarrow \mathbb{R}$,*

$$\mathbb{E}_{P^\kappa}[f(Y_1, \dots, Y_n)] = \int_{\mathcal{Y}} \cdots \int_{\mathcal{Y}} f(y_1, \dots, y_n) \kappa_{n-1}(y_{1:n-1}, dy_n) \cdots \kappa_1(y_1, dy_2) \kappa_0(\emptyset, dy_1).$$

Moreover, for every $t \geq 0$, $\kappa_t(Y_{1:t}, \cdot)$ is a version of the regular conditional law of Y_{t+1} given $\mathcal{F}_t$ under P^κ .

The proof is deferred to Appendix A. When the kernels κ_t are Borel measurable, this is the usual Ionescu–Tulcea construction; see, e.g., Kallenberg (2021). For universally measurable selectors, the construction proceeds by replacing each kernel by a Borel version under the appropriate finite-dimensional marginal.

With the path law associated to each selection fixed, we can define the composite null generated by the local model.

Definition 2.3 (Analytic sequential hypothesis generated by $\mathcal{K}$). *The analytic sequential hypothesis generated by $\mathcal{K}$ is*

$$\mathfrak{P}^{\mathcal{K}} := \{P^{\kappa} : \kappa \text{ is a } \mathcal{K}\text{-selection}\}.$$

When $\mathcal{K}$ is clear from context, we simply write $\mathfrak{P}$.

Thus $\mathfrak{P}^{\mathcal{K}}$ contains every path law obtained by selecting, at each finite history, an admissible next-step distribution from $\mathcal{K}(s)$ in a universally measurable way. In particular, admissible conditional laws may be recombined across histories without any additional global compatibility restriction beyond measurability.

A simple local model that we will return to throughout the paper is obtained by taking $\mathcal{Y} = [0, 1]$, fixing $\mu \in (0, 1)$, and setting

$$\mathcal{K}_{\mu}(s) := \left\{ Q \in \Delta([0, 1]) : \int y Q(dy) = \mu \right\}$$

for every history s . The map $Q \mapsto \int y Q(dy)$ is continuous for weak convergence, because $y \mapsto y$ is bounded and continuous on $[0, 1]$. Hence $\mathcal{K}_{\mu}(s)$ is closed in the compact space $\Delta([0, 1])$, and is therefore compact; its graph is in fact Borel. A $\mathcal{K}_{\mu}$ -selection may nevertheless choose a different mean- μ distribution after every history. Thus $\mathfrak{P}^{\mathcal{K}_{\mu}}$ is the family of path laws satisfying

$$\mathbb{E}_P[Y_{t+1} \mid \mathcal{F}_t] = \mu \quad P\text{-almost surely for every } t,$$

with no independence or stationarity requirement on the remaining features of the conditional law. This is the bounded-mean model from Example 6.1, expressed in the local language of Definition 2.3.

This local-to-global construction is the product structure behind the analytic sequential hypotheses studied here. Closely related product and pasting ideas appear under the names *rectangularity* and *analytic product structure* in robust control, multiple-priors models, and nondominated stochastic analysis (Epstein and Schneider, 2003; Iyengar, 2005; Nutz and van Handel, 2013; Bouchard and Nutz, 2015). The connection is discussed briefly in Section 1.1.

The local specification itself is pointwise in the history. A $\mathcal{K}$ -selection satisfies $\kappa_t(s) \in \mathcal{K}(s)$ at every $s \in \mathcal{Y}^t$, including histories that may have probability zero under the induced law P^{κ} . Once a particular path law is fixed, however, the identification of $\kappa_t(Y_{1:t}, \cdot)$ with a regular conditional distribution is necessarily only P^{κ} -almost sure.

2.3 E-processes and test supermartingales

We now introduce the two classes of sequential testing processes that we compare. Our terminology follows the modern anytime-valid testing literature, in which e-processes encode validity under optional stopping and test martingales or supermartingales provide a betting-based sufficient construction (Shafer et al., 2011; Ramdas et al., 2023; Grünwald et al., 2024). Let $\mathcal{T}_b$ denote the collection of bounded stopping times with respect to the canonical filtration $(\mathcal{F}_t)_{t \geq 0}$. For a Borel-adapted process E and $\tau \in \mathcal{T}_b$, we write E_{τ} for the corresponding stopped process.

Definition 2.4 (e-process). *Let $\mathfrak{P} \subseteq \Delta(\Omega)$ be a composite null. A nonnegative Borel-adapted process $E = (E_t)_{t \geq 0}$ is a $\mathfrak{P}$ -e-process if $E_0 \leq 1$ and*

$$\mathbb{E}_P[E_\tau] \leq 1 \quad \text{for every } P \in \mathfrak{P} \text{ and every } \tau \in \mathcal{T}_b. \quad (1)$$

Thus an e-process remains valid under optional stopping uniformly over the composite null. An analyst may decide when to stop on the basis of the observations seen so far, while the expected stopped value remains bounded by one under every $P \in \mathfrak{P}$.

For the nonnegative processes considered here, restricting attention to bounded stopping times is without loss. Indeed, if τ is an arbitrary stopping time, then $\tau \wedge n$ is bounded; with the convention $E_\infty := \liminf_{t \rightarrow \infty} E_t$, Fatou's lemma extends the same inequality to τ . We use bounded stopping times in the basic setup to avoid introducing additional conventions at infinity before they are needed.

Test supermartingales satisfy the stronger requirement that the process have nonpositive conditional drift under every law in the null. For $P \in \Delta(\Omega)$, let $\mathcal{F}_t^P$ denote the P -completion of $\mathcal{F}_t$.

Definition 2.5 (Test supermartingale). *Let $\mathfrak{P} \subseteq \Delta(\Omega)$ be a composite null. A nonnegative Borel-adapted process $M = (M_t)_{t \geq 0}$ is a $\mathfrak{P}$ -test supermartingale if $M_0 \leq 1$ and, for every $P \in \mathfrak{P}$ and $t \geq 0$,*

$$\mathbb{E}_P[M_{t+1} \mid \mathcal{F}_t^P] \leq M_t \quad P\text{-almost surely.} \quad (2)$$

The same process M must therefore satisfy (2) under every law in the composite null. Every $\mathfrak{P}$ -test supermartingale is a $\mathfrak{P}$ -e-process: by nonnegative optional sampling, $\mathbb{E}_P[M_\tau] \leq M_0 \leq 1$ for every $P \in \mathfrak{P}$ and every bounded stopping time τ (Kallenberg, 2021, see, e.g.,). The converse need not hold. An e-process is constrained through its expected value at stopping times, but need not itself satisfy a one-step supermartingale inequality. A particularly sharp separation occurs for testing exchangeability, where nontrivial test supermartingales can be powerless even though powerful e-processes exist (Ramdas et al., 2022).

The question studied in this paper is whether this distinction matters for testing. Although e-processes form the more general class, it may still be possible to replace every valid e-process by a valid test supermartingale that is at least as large after every possible history.

2.4 Pointwise domination and the complete-class property

We formalize this comparison using a pathwise notion of domination.

Definition 2.6 (Pointwise domination). *Let $E = (E_t)_{t \geq 0}$ and $M = (M_t)_{t \geq 0}$ be adapted processes. We say that M pointwise dominates, or superreplicates, E if*

$$M_t(s) \geq E_t(s) \quad \text{for every } t \geq 0 \text{ and every } s \in \mathcal{Y}^t. \quad (3)$$

The historywise requirement (3) is stronger than requiring domination P -almost surely for every $P \in \mathfrak{P}$. The inequality must hold at every finite history, including histories that are null under some or all laws in the composite null. This pathwise order induces the complete-class notion in Definition 2.7.

Definition 2.7 (Complete class). *Let $\mathfrak{P} \subseteq \Delta(\Omega)$. We say that test supermartingales form a complete class for $\mathfrak{P}$ if every $\mathfrak{P}$ -e-process E is pointwise dominated by some $\mathfrak{P}$ -test supermartingale M .*

The complete-class property gives a direct testing interpretation. Fix a confidence-level $\alpha \in (0, 1)$ and let

$$\tau_E := \inf \left\{ t \geq 0 : E_t \geq \frac{1}{\alpha} \right\}$$

be the rejection time obtained by thresholding an e-process. If M pointwise dominates E and τ_M denotes the analogous rejection time for M , then

$$\tau_M \leq \tau_E$$

along every path. Thus any rejection made by E is made no later by M . In this sense, pointwise domination shows that restricting attention to test supermartingales entails no loss of anytime-valid testing power.

Theorem 5.1 establishes this complete-class property for the analytic sequential hypotheses $\mathfrak{P}^{\mathcal{K}}$ generated by Definition 2.3.

Remark 2.8 (Measurability convention). The testing objects in Definitions 2.4 and 2.5 are Borel adapted. The analytic-graph assumption on $\mathcal{K}$, however, naturally leads to universally measurable selectors and, in the Snell-envelope construction used in the proof, to upper semianalytic and universally adapted intermediate processes. We introduce these larger measurability classes only when they become necessary. Section 4, culminating in Proposition 4.3, passes from this universally measurable intermediate construction back to a Borel-adapted dominating test supermartingale.

3 Constructing the Snell envelope

The first step in the proof is to show that an e-process can be enlarged to a process satisfying the supermartingale inequalities pointwise at every history. The natural construction is a robust Snell envelope. For a fixed probability law, the classical Snell envelope is the smallest supermartingale dominating a reward process and is the standard dynamic-programming object for optimal stopping (Peskir and Shiryaev, 2006; Shiryaev, 2008). A composite version also appears in Ramdas et al. (2022, Theorem 14): for a locally dominated, fork-convex family of laws, they construct the smallest composite nonnegative supermartingale dominating a safe e-process by taking essential suprema of conditional stopped rewards relative to a common reference measure.

Our construction serves the same basic purpose—turning optional-stopping safety into dynamic supermartingale inequalities—but is organized differently. The null is given locally through the historywise sets $\mathcal{K}(s)$, so the recursion is performed pointwise on the history space using a robust one-step expectation. No common locally dominating reference measure is required, and we do not assume fork-convexity of the global family. In return, the natural Bellman operator preserves upper semianalyticity rather than Borel measurability, so a single Snell step may leave the Borel class. We also do not need to identify an almost-surely smallest composite envelope: for the complete-class theorem, it is enough to construct a pointwise dominating process satisfying the local inequalities with initial capital at most one.

Because the one-step optimization may leave the Borel class, we first enlarge the measurability class in which the intermediate recursion is carried out. A function f on a Polish space is called *upper semianalytic* (**usa**) if $\{f > a\}$ is analytic for every $a \in \mathbb{R}$. Every **usa** function is universally measurable; see, e.g., Bertsekas and Shreve (1996, Chapter 7) for these standard analytic-measurability facts. We call a nonnegative process $X = (X_t)_{t \geq 0}$ a ***usa** process* if each history-space representative $X_t : \mathcal{Y}^t \rightarrow [0, \infty]$ is **usa**. A bounded stopping time τ is called

universally measurable if, for every t , the event $\{\tau \leq t\}$ can be written as $\{Y_{1:t} \in A_t\}$ for some $A_t \in \mathcal{U}_t$. We write $\mathcal{T}_b^u$ for the class of bounded universally measurable stopping times.

For the intermediate arguments in Section 3, we say that a nonnegative **usa** process E is a **usa** $\mathfrak{P}^\kappa$ -e-process if $E_0 \leq 1$ and $\mathbb{E}_{P^\kappa}[E_\tau] \leq 1$ for every $\mathcal{K}$ -selection κ and every $\tau \in \mathcal{T}_b^u$. Lemma 3.1 lets us pass from the Borel definition in Section 2 to this larger class.

Lemma 3.1 (Borel e-processes are valid at universal stopping times). *Every Borel $\mathfrak{P}^\kappa$ -e-process is a **usa** $\mathfrak{P}^\kappa$ -e-process.*

Proof. Fix a $\mathcal{K}$ -selection κ and $\tau \in \mathcal{T}_b^u$. By Lemma B.1, there is a bounded Borel stopping time σ such that $\sigma = \tau$ P^κ -almost surely. Since E is Borel adapted, $E_\sigma = E_\tau$ P^κ -almost surely, and Definition 2.4 gives $\mathbb{E}_{P^\kappa}[E_\tau] = \mathbb{E}_{P^\kappa}[E_\sigma] \leq 1$. Every Borel function is **usa**, so the claim follows. $\square$

3.1 The one-step upper expectation and finite-horizon recursion

For a nonnegative **usa** function $f : \mathcal{Y}^{t+1} \rightarrow [0, \infty]$, define the *one-step upper expectation* $\mathbb{T}_t f : \mathcal{Y}^t \rightarrow [0, \infty]$ by

$$(\mathbb{T}_t f)(s) := \sup_{Q \in \mathcal{K}(s)} \int_{\mathcal{Y}} f(s \oplus y) Q(dy). \quad (4)$$

Thus the quantity in (4) is the largest expected continuation value compatible with the local model after history s . The analytic-graph assumption ensures that this operation preserves the measurability class needed for the recursion.

Lemma 3.2 (Analytic dynamic-programming facts). *Let $f, f_n : \mathcal{Y}^{t+1} \rightarrow [0, \infty]$ be **usa**. Then $\mathbb{T}_t f$ is **usa**, $\mathbb{T}_t$ is monotone, and if $f_n \uparrow f$ pointwise then $\mathbb{T}_t f_n \uparrow \mathbb{T}_t f$ pointwise. Moreover, if f is bounded, then for every $\delta > 0$ there is a universally measurable selector $\kappa_t^\delta : \mathcal{Y}^t \rightarrow \Delta(\mathcal{Y})$ satisfying $\kappa_t^\delta(s) \in \mathcal{K}(s)$ and*

$$\int_{\mathcal{Y}} f(s \oplus y) \kappa_t^\delta(s, dy) \geq (\mathbb{T}_t f)(s) - \delta$$

for every $s \in \mathcal{Y}^t$.

The proof is standard analytic dynamic programming; see, e.g., Bertsekas and Shreve (1996, Chapter 7) and related nondominated dynamic-programming constructions in Nutz and van Handel (2013); Bouchard and Nutz (2015). We include the details in Appendix C for completeness. The final statement is the only place where a maximizing conditional law is needed; importantly, the supremum defining $\mathbb{T}_t$ need not be attained, so the selector is only approximately maximizing.

For the bounded-mean model $\mathcal{K}_\mu$ from Example 6.1, fix a history s and write $f_s(y) := f(s \oplus y)$. If f_s is bounded and continuous, then

$$(\mathbb{T}_t f)(s) = \sup_{Q: \mathbb{E}_Q Y = \mu} \mathbb{E}_Q[f_s(Y)] = \text{conc}(f_s)(\mu),$$

where $\text{conc}(f_s)$ denotes the least concave majorant of f_s on $[0, 1]$. Indeed, every concave majorant $g \geq f_s$ gives

$$\mathbb{E}_Q[f_s(Y)] \leq \mathbb{E}_Q[g(Y)] \leq g(\mathbb{E}_Q Y) = g(\mu),$$

Protocol 1: Finite-horizon robust Snell envelope

Input: A horizon $N \in \mathbb{N}$ and a nonnegative **usa** process $E = (E_t)_{t=0}^N$.

Output: The finite-horizon robust Snell envelope $M^{(N)} = (M_t^{(N)})_{t=0}^N$.

Set $M_N^{(N)} := E_N$.

for $t = N - 1, \dots, 0$ **do**

 Define $M_t^{(N)} : \mathcal{Y}^t \rightarrow [0, \infty]$ by

$$M_t^{(N)}(s) := \max \left\{ \underbrace{E_t(s)}_{\text{stop}}, \underbrace{(\mathsf{T}_t M_{t+1}^{(N)})(s)}_{\text{continue}} \right\}, \quad s \in \mathcal{Y}^t.$$

and in one dimension the bound is attained, or approximated arbitrarily well, by a distribution supported on at most two points whose barycenter is μ .

When f is jointly continuous, compactness of $\mathcal{K}_\mu$ and the maximum theorem show that $s \mapsto (\mathsf{T}_t f)(s)$ is continuous. Finite-horizon Snell envelopes of continuous reward processes are therefore continuous, and in particular Borel. This regularity should not be confused with a general Borel-preservation statement: for an arbitrary Borel continuation payoff, or for a general analytic local model, the robust supremum need only be upper semianalytic. The larger measurability class used in Section 3 is needed precisely for that general case. The familiar bounded-mean betting martingales also fit the recursion directly. If

$$M_{t+1}(s \oplus y) = M_t(s)(1 + \lambda_t(s)(y - \mu))$$

with a predictable stake chosen so that the factor is nonnegative, then

$$(\mathsf{T}_t M_{t+1})(s) = M_t(s).$$

Thus these classical betting processes are fixed points of the one-step Snell operation; the Snell envelope of a general e-process need not itself have this affine betting form.

With the one-step operator in hand, we now build the envelope over a finite horizon. Fix a finite horizon $N \in \mathbb{N}$ and a nonnegative **usa** process E . The finite-horizon robust Snell envelope is obtained by the backward recursion in Protocol 1. At each history, the envelope compares the value $E_t(s)$ obtained by stopping immediately with the largest expected value of continuing for one more step under an admissible conditional law.

By Lemma 3.2, every $M_t^{(N)}$ produced by Protocol 1 is **usa**. Moreover, the two defining properties of the envelope are immediate from the recursion:

$$M_t^{(N)} \geq E_t, \quad \mathsf{T}_t M_{t+1}^{(N)} \leq M_t^{(N)} \tag{5}$$

pointwise on $\mathcal{Y}^t$. The first inequality in (5) gives domination of the original process, while the second is the robust one-step supermartingale inequality. The remaining issue is to show that enforcing (5) does not require more than the unit initial-capital budget available to an e-process. Proposition 3.3 establishes precisely this capital bound.

3.2 Finite-horizon capital preservation

The key argument is most transparent for bounded rewards. The envelope prescribes stopping when it first meets the reward, while measurable selection supplies conditional laws that come arbitrarily close to realizing the worst-case continuation value.

Proposition 3.3 (Finite-horizon capital bound). *Let E be a **usa** $\mathfrak{P}^{\mathcal{K}}$ -e-process, and let $M^{(N)}$ be its finite-horizon robust Snell envelope. Then $M_0^{(N)} \leq 1$ for every $N \in \mathbb{N}$. More generally, if a nonnegative **usa** process E satisfies*

$$\mathbb{E}_{P^\kappa}[E_\tau] \leq c$$

for every $\mathcal{K}$ -selection κ and every $\tau \in \mathcal{T}_b^u$, then $M_0^{(N)} \leq c$ for every N .

Proof. We first suppose that $E_0, \dots, E_N$ are bounded by a common constant $L < \infty$. Then the recursion and preservation of constants by $\mathbb{T}_t$ imply $0 \leq M_t^{(N)} \leq L$ for every $t \leq N$. Fix $\varepsilon > 0$, and choose $\delta_0, \dots, \delta_{N-1} > 0$ with $\sum_{t=0}^{N-1} \delta_t < \varepsilon$. For each $t < N$, Lemma 3.2 gives a universally measurable selector κ_t^ε such that

$$\int_{\mathcal{Y}} M_{t+1}^{(N)}(s \oplus y) \kappa_t^\varepsilon(s, dy) \geq (\mathbb{T}_t M_{t+1}^{(N)})(s) - \delta_t$$

for every $s \in \mathcal{Y}^t$. Complete these kernels after time N using any universally measurable selector of $\mathcal{K}$, and let P^{κ^ε} be the induced path law.

Define the first contact time

$$\tau^* := \inf\{0 \leq t \leq N : M_t^{(N)}(Y_{1:t}) = E_t(Y_{1:t})\}.$$

Since $M_N^{(N)} = E_N$, this stopping time is finite, and Lemma B.3 gives $\tau^* \in \mathcal{T}_b^u$. On $\{t < \tau^*\}$ we have $M_t^{(N)} > E_t$, and hence the Snell recursion gives $M_t^{(N)} = \mathbb{T}_t M_{t+1}^{(N)}$. Using Lemma B.2 and the choice of κ_t^ε , we therefore obtain

$$\mathbb{E}_{P^{\kappa^\varepsilon}} \left[M_{t+1}^{(N)}(Y_{1:t+1}) \mid \mathcal{F}_t^{P^{\kappa^\varepsilon}} \right] \geq M_t^{(N)}(Y_{1:t}) - \delta_t$$

on $\{t < \tau^*\}$, P^{κ^ε} -almost surely. Consequently,

$$Z_t := M_{t \wedge \tau^*}^{(N)}(Y_{1:t \wedge \tau^*}) + \sum_{j=0}^{(t \wedge \tau^*)-1} \delta_j, \quad 0 \leq t \leq N,$$

is a bounded P^{κ^ε} -submartingale. At the stopping time, exact contact gives $M_{\tau^*}^{(N)} = E_{\tau^*}$, and hence

$$M_0^{(N)} = Z_0 \leq \mathbb{E}_{P^{\kappa^\varepsilon}}[Z_N] \leq \mathbb{E}_{P^{\kappa^\varepsilon}}[E_{\tau^*}] + \sum_{t=0}^{N-1} \delta_t < 1 + \varepsilon.$$

The e-process inequality applies because τ^* is a bounded universally measurable stopping time. Letting $\varepsilon \downarrow 0$ yields $M_0^{(N)} \leq 1$ in the bounded case.

For a general nonnegative E , set $E_t^{(m)} := E_t \wedge m$ and let $M^{(N,m)}$ be the corresponding finite-horizon envelope. Each $E^{(m)}$ is again a **usa** $\mathfrak{P}^{\mathcal{K}}$ -e-process, so the bounded argument gives $M_0^{(N,m)} \leq 1$. By continuity from below of $\mathbb{T}_t$ and backward induction, $M_t^{(N,m)} \uparrow M_t^{(N)}$ for every $t \leq N$. Therefore $M_0^{(N)} = \sup_m M_0^{(N,m)} \leq 1$. The general bound with right-hand side c follows from the same argument, with the e-process bound 1 replaced throughout by c . $\square$

3.3 Infinite-horizon Snell domination

We now let the stopping horizon increase. For each $t \geq 0$, define

$$M_t := \sup_{N \geq t} M_t^{(N)},$$

where the supremum is pointwise on $\mathcal{Y}^t$. The finite-horizon envelopes are monotone in the horizon: for fixed t , $M_t^{(N)} \leq M_t^{(N+1)}$ whenever $N \geq t$. Indeed, $M_N^{(N+1)} = E_N \vee \mathsf{T}_N E_{N+1} \geq E_N = M_N^{(N)}$, and monotonicity of the one-step operators propagates this inequality backwards. The resulting monotone limit retains the domination, capital, and one-step inequalities established at finite horizons.

Theorem 3.4 (*usa Snell domination*). *Let E be a **usa** $\mathfrak{P}^\kappa$ -e-process. Then the process $M = (M_t)_{t \geq 0}$ defined by the increasing-horizon limit in Section 3.3 is **usa**, satisfies $M_t \geq E_t$ pointwise for every t , has $M_0 \leq 1$, and obeys*

$$\mathsf{T}_t M_{t+1}(s) \leq M_t(s) \quad \text{for every } t \geq 0 \text{ and } s \in \mathcal{Y}^t.$$

Consequently, for every $\mathcal{K}$ -selection κ , the canonical process $M_t(Y_{1:t})$ is a nonnegative P^κ -supermartingale with respect to the P^κ -completed canonical filtration.

Proof. For each fixed t , M_t is the increasing countable supremum of the **usa** functions $M_t^{(N)}$, and is therefore **usa**. Moreover, $M_t \geq M_t^{(t)} = E_t$ pointwise, while Proposition 3.3 gives $M_0 = \sup_N M_0^{(N)} \leq 1$. Finally, continuity from below of T_t yields

$$\mathsf{T}_t M_{t+1} = \sup_{N \geq t+1} \mathsf{T}_t M_{t+1}^{(N)} \leq \sup_{N \geq t+1} M_t^{(N)} \leq M_t.$$

Fixing a $\mathcal{K}$ -selection κ , Lemma B.2 converts this pointwise kernel inequality into

$$\mathbb{E}_{P^\kappa} [M_{t+1}(Y_{1:t+1}) \mid \mathcal{F}_t^{P^\kappa}] \leq M_t(Y_{1:t}) \quad P^\kappa\text{-almost surely,}$$

which is the required supermartingale property. $\square$

Theorem 3.4 is the first domination step in the proof of our main result. Starting from a Borel $\mathfrak{P}^\kappa$ -e-process, Lemma 3.1 lets us apply the theorem and obtain a pointwise dominating supermartingale with no more than unit initial capital. The construction also satisfies the stronger one-step inequalities at every finite history. What it does not yet provide is Borel measurability: even when E is Borel, the operator T_t need only preserve upper semianalyticity. Section 4 addresses precisely this remaining obstruction.

4 From universal measurability to Borel supermartingales

The Snell construction of Section 3 produces the desired pointwise one-step inequalities, but generally only within the upper-semianalytic class. A second operation is therefore needed to return to Borel processes. The key observation is that a **usa** process which is valid at its own initial capital can be enlarged pointwise to a Borel process with the same optional-stopping guarantee and the same initial capital. This reflection step need not preserve the one-step supermartingale inequalities, so we alternate it with the Snell construction and then pass to a monotone limit.

The argument produces two interlaced sequences. The reflected processes provide Borel adaptedness, while the intervening Snell envelopes provide the stronger pointwise supermartingale inequalities. Neither sequence alone has both properties after a single step. The monotone interlacing is what allows the limit to inherit Borel measurability from the reflected sequence and the local supermartingale structure from the Snell sequence.

4.1 Reflecting a *usa* process into the Borel class

The reflection step is most conveniently formulated at the level of analytic subgraphs. For an analytic set $D \subseteq \mathbf{H} \times [0, \infty)$, write $D_s := \{u \geq 0 : (s, u) \in D\}$ and let $g_D(s) := \lambda(D_s)$, where λ denotes Lebesgue measure on $[0, \infty)$. Define its *robust stopped-value cost* by

$$\mathcal{C}(D) := \sup_{\kappa} \sup_{\tau \in \mathcal{T}_{\mathbf{b}}^{\mathbf{u}}} \mathbb{E}_{P^{\kappa}}[g_D(Y_{1:\tau})],$$

where the first supremum ranges over all $\mathcal{K}$ -selections. The functional $\mathcal{C}$ is monotone under set inclusion. The descriptive-set-theoretic input for reflection is that an analytic set can be enlarged to a Borel set without increasing this cost.

Proposition 4.1 (Borel hull preserving stopped-value cost). *Let $D \subseteq \mathbf{H} \times [0, \infty)$ be analytic with $\mathcal{C}(D) < \infty$. Then there exists a Borel set $B \subseteq \mathbf{H} \times [0, \infty)$ such that $D \subseteq B$ and $\mathcal{C}(B) = \mathcal{C}(D)$.*

The proof is given in Appendix D. It combines a parameterized version of the finite-horizon dynamic-programming argument from Section 3 with the dual First Reflection Theorem from descriptive set theory (Kechris, 1995, Theorem 35.10 and the discussion following it). We use Proposition 4.1 through its process-level consequence, Proposition 4.2.

Proposition 4.2 (Process reflection). *Let $X = (X_t)_{t \geq 0}$ be a nonnegative *usa* process with $X_0 < \infty$, and suppose that*

$$\mathbb{E}_{P^{\kappa}}[X_{\tau}] \leq X_0$$

for every $\mathcal{K}$ -selection κ and every $\tau \in \mathcal{T}_{\mathbf{b}}^{\mathbf{u}}$. Then there exists a nonnegative Borel-adapted process $\mathbf{R}(X)$ such that $\mathbf{R}(X)_t(s) \geq X_t(s)$ for every t and $s \in \mathcal{Y}^t$, $\mathbf{R}(X)_0 = X_0$, and

$$\mathbb{E}_{P^{\kappa}}[\mathbf{R}(X)_{\tau}] \leq \mathbf{R}(X)_0$$

for every $\mathcal{K}$ -selection κ and every $\tau \in \mathcal{T}_{\mathbf{b}}^{\mathbf{u}}$.

Proof. View X as a function on the disjoint-union history space $\mathbf{H}$ and define its strict subgraph

$$D_X := \{(s, u) \in \mathbf{H} \times [0, \infty) : 0 \leq u < X(s)\}.$$

Since X is *usa*, D_X is analytic, and its section length satisfies $g_{D_X}(s) = X(s)$ for every history s . The assumed optional-stopping bound gives $\mathcal{C}(D_X) \leq X_0$, while the deterministic stopping time $\tau \equiv 0$ gives the reverse inequality. Hence $\mathcal{C}(D_X) = X_0$.

By Proposition 4.1, there is a Borel set $B \supseteq D_X$ with $\mathcal{C}(B) = X_0$. Define $\mathbf{R}(X)(s) := \lambda(B_s)$. Because B is Borel, $\mathbf{R}(X)$ is Borel adapted, and $B \supseteq D_X$ gives $\mathbf{R}(X) \geq X$ pointwise. Moreover, stopping at time zero gives $\mathbf{R}(X)_0 \leq \mathcal{C}(B) = X_0$, while pointwise domination gives $\mathbf{R}(X)_0 \geq X_0$. Thus $\mathbf{R}(X)_0 = X_0$. Finally, the definition of $\mathcal{C}(B)$ gives $\mathbb{E}_{P^{\kappa}}[\mathbf{R}(X)_{\tau}] \leq \mathcal{C}(B) = \mathbf{R}(X)_0$ for every κ and every $\tau \in \mathcal{T}_{\mathbf{b}}^{\mathbf{u}}$. $\square$

The reflection proposition applies in particular to the output of the Snell construction. Indeed, if M is a nonnegative **usa** process with $M_0 < \infty$ satisfying $\mathsf{T}_t M_{t+1} \leq M_t$ pointwise, then Theorem 3.4 and bounded optional sampling imply

$$\mathbb{E}_{P^\kappa}[M_\tau] \leq M_0$$

for every $\mathcal{K}$ -selection κ and every $\tau \in \mathcal{T}_b^u$. Thus reflection restores Borel measurability without increasing the capital required by a **usa** Snell envelope. It does not, however, necessarily preserve the local inequalities $\mathsf{T}_t M_{t+1} \leq M_t$. This is why one further Snell step is required.

4.2 Alternating Snell and reflection

We now combine the two complementary operations from Theorem 3.4 and Proposition 4.2. Starting from a Borel e -process, the Snell construction produces a dominating **usa** process satisfying the pointwise one-step inequalities; reflection then produces a still larger Borel process which is valid at its own initial capital. Repeating this procedure produces the increasing sequence formalized in Proposition 4.3, whose limit is simultaneously Borel and a pointwise supermartingale.

Proposition 4.3 (Alternating Snell–reflection construction). *Let E be a Borel $\mathfrak{P}^\mathcal{K}$ - e -process. Then there exists a nonnegative Borel-adapted process $G = (G_t)_{t \geq 0}$ such that $G_0 \leq 1$, $G_t(s) \geq E_t(s)$ for every $t \geq 0$ and $s \in \mathcal{Y}^t$, and*

$$\mathsf{T}_t G_{t+1}(s) \leq G_t(s) \quad \text{for every } t \geq 0 \text{ and } s \in \mathcal{Y}^t.$$

Consequently, G is a $\mathfrak{P}^\mathcal{K}$ -test supermartingale which pointwise dominates E .

Proof. Set $E^{(0)} := E$ and apply Theorem 3.4 to obtain a **usa** Snell envelope $M^{(0)}$ satisfying $E^{(0)} \leq M^{(0)}$ pointwise, $\mathsf{T}_t M_{t+1}^{(0)} \leq M_t^{(0)}$ for every t , and $M_0^{(0)} \leq 1$. Write $c := M_0^{(0)}$. By the supermartingale consequence of Theorem 3.4 and bounded optional sampling, $M^{(0)}$ is valid under bounded universal stopping at its own initial capital c , so Proposition 4.2 gives a Borel process $E^{(1)} := \mathsf{R}(M^{(0)})$ with $M^{(0)} \leq E^{(1)}$, $E_0^{(1)} = c$, and $\mathbb{E}_{P^\kappa}[E_\tau^{(1)}] \leq c$ for every κ and $\tau \in \mathcal{T}_b^u$.

Proceed inductively. Suppose $E^{(n)}$ is Borel, $E_0^{(n)} = c$, and is valid at its own initial capital c . Apply Theorem 3.4, using the scaled capital bound from Proposition 3.3. By the scaled form of Proposition 3.3, its **usa** Snell envelope $M^{(n)}$ satisfies

$$E^{(n)} \leq M^{(n)}, \quad \mathsf{T}_t M_{t+1}^{(n)} \leq M_t^{(n)}, \quad M_0^{(n)} = c.$$

Reflecting once more, define $E^{(n+1)} := \mathsf{R}(M^{(n)})$. Then $E^{(n+1)}$ is Borel, $M^{(n)} \leq E^{(n+1)}$ pointwise, $E_0^{(n+1)} = c$, and $E^{(n+1)}$ is again valid at its own initial capital. Thus the processes interlace pointwise as

$$E^{(0)} \leq M^{(0)} \leq E^{(1)} \leq M^{(1)} \leq E^{(2)} \leq \dots$$

For each $t \geq 0$ and $s \in \mathcal{Y}^t$, define

$$G_t(s) := \sup_{n \geq 0} E_t^{(n)}(s).$$

Every $E_t^{(n)}$ is Borel, so G_t is Borel as a countable supremum. The interlacing gives $G \geq E^{(0)} = E$ pointwise, and $G_0 = c \leq 1$. It remains to verify the one-step inequality. For every n and t ,

$$E_t^{(n+1)} \geq M_t^{(n)} \geq \mathsf{T}_t M_{t+1}^{(n)} \geq \mathsf{T}_t E_{t+1}^{(n)},$$

where the final inequality uses monotonicity of T_t . Since $E_t^{(n)} \uparrow G_t$ pointwise, continuity from below of T_t gives

$$G_t = \sup_{n \geq 0} E_t^{(n+1)} \geq \sup_{n \geq 0} \mathsf{T}_t E_{t+1}^{(n)} = \mathsf{T}_t G_{t+1}.$$

Thus $\mathsf{T}_t G_{t+1}(s) \leq G_t(s)$ at every finite history. Finally, Lemma B.2 applied to the Borel process G shows that, under every $P^\kappa \in \mathfrak{P}^\mathcal{K}$,

$$\mathbb{E}_{P^\kappa} [G_{t+1}(Y_{1:t+1}) \mid \mathcal{F}_t^{P^\kappa}] \leq G_t(Y_{1:t}) \quad P^\kappa\text{-almost surely.}$$

Hence G is a $\mathfrak{P}^\mathcal{K}$ -test supermartingale. $\square$

Proposition 4.3 resolves the measurability obstruction left by Section 3. More concretely, the final process is defined as a countable supremum of Borel reflected processes, which gives its Borel adaptedness, while the inequalities inherited through the interlaced Snell envelopes and continuity from below of T_t give its pointwise supermartingale property. Thus the two alternating operations contribute genuinely different parts of the final object. The argument is existential: measurable selection and reflection guarantee the required intermediate objects, but do not in general provide an algorithm for computing the dominating test supermartingale.

5 Complete-class theorem

The **usa** Snell domination theorem (Theorem 3.4) and the Snell–reflection construction (Proposition 4.3) now give the main result. The assumptions are exactly those in Section 2: $\mathcal{K}$ is nonempty-valued with analytic graph, and $\mathfrak{P}^\mathcal{K}$ contains all path laws generated by universally measurable $\mathcal{K}$ -selections.

Theorem 5.1 (Complete class for analytic sequential hypotheses). *Let $\mathcal{K} : \mathsf{H} \rightrightarrows \Delta(\mathcal{Y})$ be an analytic local model, and let $\mathfrak{P}^\mathcal{K}$ be the analytic sequential hypothesis it generates. Then test supermartingales form a complete class for $\mathfrak{P}^\mathcal{K}$ in the sense of Definition 2.7. More precisely, for every $\mathfrak{P}^\mathcal{K}$ -e-process $E = (E_t)_{t \geq 0}$ there exists a nonnegative Borel-adapted process $G = (G_t)_{t \geq 0}$ with $G_0 \leq 1$ such that*

$$G_t(s) \geq E_t(s) \quad \text{for every } t \geq 0 \text{ and } s \in \mathcal{Y}^t, \quad (6)$$

and

$$\sup_{Q \in \mathcal{K}(s)} \int_{\mathcal{Y}} G_{t+1}(s \oplus y) Q(dy) \leq G_t(s) \quad \text{for every } t \geq 0 \text{ and } s \in \mathcal{Y}^t. \quad (7)$$

In particular, G is a $\mathfrak{P}^\mathcal{K}$ -test supermartingale that pointwise dominates E .

Proof. Proposition 4.3 gives a nonnegative Borel process G with $G_0 \leq 1$, satisfying the pointwise domination in (6) and the one-step inequalities in (7). Under any $P^\kappa \in \mathfrak{P}^\mathcal{K}$, the selected kernel $\kappa_t(Y_{1:t}, \cdot)$ is a version of the conditional law of Y_{t+1} given the past. Lemma B.2 therefore turns the pointwise kernel inequality into

$$\mathbb{E}_{P^\kappa} [G_{t+1}(Y_{1:t+1}) \mid \mathcal{F}_t^{P^\kappa}] \leq G_t(Y_{1:t}) \quad P^\kappa\text{-almost surely.}$$

Thus G is a test supermartingale for every law in $\mathfrak{P}^\mathcal{K}$. $\square$

Theorem 5.1 yields slightly more than the complete-class formulation alone records. The historywise inequality (7) is stronger than the complete-class statement itself. It holds after every finite history, including histories that may be null under some or all laws in the hypothesis. Accordingly, if an e-process crosses a rejection threshold on a given path, its dominating test supermartingale crosses no later.

6 Examples

We consider five examples. Bounded conditional means, introduced in Example 6.1, provide a familiar reference point. The conditional-FOSD, sub-Gaussian, and Wasserstein examples in Sections 6.3–6.5 impose distributional or tail restrictions that are statistically standard but are no longer described by a single conditional moment. The latent-feasibility example in Section 6.6 is motivated by latent-variable and partial-identification problems in which admissibility is determined by the existence of an unobserved feasible object. Such existential constraints arise by projection and can naturally lead from Borel relations to analytic, non-Borel compatibility sets (Kline and Tamer, 2016; Beresteanu et al., 2012; Gu and Russell, 2023). The construction in Section 6.6 gives a simple sequential version of this mechanism and exhibits the role of analytic, rather than merely Borel, measurability in Theorem 5.1.

Two structural checks recur throughout Sections 6.1–6.6. First, many of the local classes are *not locally dominated*: there is no σ -finite measure that dominates every law in a local set $\mathcal{K}(s)$. A simple sufficient condition is the presence of uncountably many admissible laws carrying positive mass on pairwise disjoint measurable sets. Second, several of our local classes cannot be represented by the finite-dimensional one-step conditional constraint model underlying Clerico (2026),

$$\mathcal{H}_{\Phi,S} = \{Q : \mathbb{E}_Q \|\Phi\|_1 < \infty, \quad \mathbb{E}_Q[\Phi] \in S\}, \quad (8)$$

with finite-dimensional Borel Φ and nonempty convex S . The common proof for FOSD (Section 6.3), sub-Gaussianity (Section 6.4), and the nondegenerate Wasserstein balls (Section 6.5) uses the one-step affine-domination result of Clerico (2026): an infinite-dimensional family of e-variables is made simultaneously tight under one null law, which would force that entire family into the finite-dimensional span of $1, \Phi_1, \dots, \Phi_d$. The analytic example is outside the same framework for a different, more immediate reason: its local correspondence has a genuinely non-Borel dependence on the history. Closely related projection-based measurability issues arise in partial-identification problems when model compatibility is defined through the existence of a latent or structural parameter satisfying a measurable feasibility relation (Kline and Tamer, 2016; Beresteanu et al., 2012; Gu and Russell, 2023). The reusable obstruction lemmas are Propositions E.1 and E.2, with the Borel-graph obstruction in Proposition E.3; all example-specific proofs are collected in Appendix E.

These comparisons clarify the scope of Theorem 5.1. The finite-dimensional theory gives a much more explicit affine parametrization when it applies, while our local formulation also covers conditional nulls for which that parametrization is unavailable, including nondominated and genuinely analytic examples.

6.1 Example 1: bounded conditional means

Example 6.1 (Bounded conditional mean). Let $\mathcal{Y} = [0, 1]$ and fix $\mu \in (0, 1)$. For every history s , set

$$\mathcal{K}_\mu(s) := \{Q \in \Delta([0, 1]) : \mathbb{E}_Q[Y] = \mu\}.$$

The induced sequential null requires

$$\mathbb{E}_P[Y_{t+1} \mid \mathcal{F}_t] = \mu$$

at every time, while allowing the rest of the conditional distribution to vary freely with time and history. The graph is Borel and the local sets are compact and convex. The class is

nevertheless not locally dominated; Appendix E.2 gives a one-line construction of uncountably many admissible laws with distinct atoms.

This is a familiar setting for betting-based sequential inference. If λ_t is predictable and chosen so that the factor is nonnegative, then

$$L_t(y) = 1 + \lambda_t(y - \mu)$$

has conditional expectation one under every admissible next-step law, and products of these factors are test martingales. Mixtures and adaptive choices of the stake give the usual bounded-mean betting constructions; see, for example, Howard et al. (2021) and related work on betting confidence sequences.

The example is also inside the finite-dimensional framework of Clerico (2026): take $\Phi(y) = y$ and $S = \{\mu\}$. It therefore serves as a familiar finite-dimensional reference point. As discussed in Section 3.1, its upper transition has the explicit form

$$(\mathsf{T}_t f)(s) = \text{conc}(f(s \oplus \cdot))(\mu)$$

for bounded continuous continuation values. For this model, the abstract Snell recursion reduces to a familiar concavification operation.

6.2 Beyond bounded means

The remaining examples are all locally nondominated. FOSD and sub-Gaussianity are also outside the finite-dimensional conditional-constraint framework without additional assumptions. For Wasserstein balls (Section 6.5), nondomination holds in the full compact-metric formulation, while the finite-dimensional question depends on the geometry of the center P_0 ; Appendix E.5 gives a simple diffuse-mass condition under which the ball is beyond the finite-dimensional framework. The final analytic example is outside that framework because its admissible one-step set changes with the history through an analytic non-Borel compatibility set arising from latent feasibility.

6.3 Example 2: conditional first-order stochastic dominance

Let $\mathcal{Y} = [0, 1]^2$ and write one observation as (X, Z) . For $Q \in \Delta([0, 1]^2)$, let Q_X and Q_Z denote the two marginals, and define

$$\mathcal{K}_{\text{FOSD}}(s) := \{Q : Q_Z \preceq_{\text{FOSD}} Q_X\}$$

for every history s . Equivalently,

$$F_{Q_X}(c) \leq F_{Q_Z}(c) \quad \text{for every } c \in [0, 1].$$

Thus the sequential null requires

$$P(X_{t+1} \leq c \mid \mathcal{F}_t) \leq P(Z_{t+1} \leq c \mid \mathcal{F}_t)$$

for every threshold c and every time t . The conditional joint law may otherwise change arbitrarily with the past, and the within-round dependence between X_{t+1} and Z_{t+1} is unrestricted.

The local set is closed, convex, and compact, so $\text{Gr}(\mathcal{K}_{\text{FOSD}})$ is Borel. It is not locally dominated, and it has no finite-dimensional representation of the form $\mathcal{H}_{\Phi, S}$ in (8). Appendix E.3

verifies both claims using Propositions E.1 and E.2. For the finite-dimensional obstruction, the useful witnesses are the threshold e-variables

$$E_c(x, z) = 1 + \eta (\mathbb{1}\{x \leq c\} - \mathbb{1}\{z \leq c\}), \quad c \in (0, 1),$$

with $0 < \eta \leq 1$. They are all tight under one null law with equal continuous marginals, and distinct thresholds give an infinite-dimensional family.

The same threshold functions also furnish natural betting strategies for this null. Recent work develops anytime-valid betting tests for stochastic dominance, including e-process constructions built from threshold information (Arnold et al., 2026). Our conditional null uses the same basic distributional geometry but allows the admissible conditional law to change with the history. Theorem 5.1 says that, for this conditional-FOSD hypothesis, every Borel e-process is pointwise dominated by a Borel test supermartingale. Thus improvements over a particular threshold strategy may require better bets or better predictable aggregation, but not a move outside the test-supermartingale class.

For regular continuation values the Snell operator is also well behaved. Because the local set is constant and compact, if $f(s \oplus y)$ is jointly continuous then

$$(\mathsf{T}_t f)(s) = \sup_{Q \in \mathcal{K}_{\text{FOSD}}} \mathbb{E}_Q[f(s \oplus Y)]$$

is continuous in s by the maximum theorem. For arbitrary Borel e-processes, however, we retain the general upper-semianalytic construction of Section 3.

6.4 Example 3: conditional sub-Gaussianity

Let $\mathcal{Y} = \mathbb{R}$ and normalize the sub-Gaussian variance proxy to one. Set, for every history s ,

$$\mathcal{K}_{\text{SG}}(s) := \left\{ Q \in \Delta(\mathbb{R}) : \mathbb{E}_Q[e^{\lambda Y}] \leq e^{\lambda^2/2} \text{ for every } \lambda \in \mathbb{R} \right\}.$$

The induced sequential null is

$$\mathbb{E}_P \left[e^{\lambda Y_{t+1}} \mid \mathcal{F}_t \right] \leq e^{\lambda^2/2} \quad \text{for every } \lambda \in \mathbb{R}$$

at every time. A predictable location-and-scale version can be written by applying the same condition to a standardized increment.

This is the usual sub-Gaussian assumption viewed as a null to be tested rather than merely as a condition used to prove concentration. The defining inequalities immediately yield the classical factors

$$L_\lambda(y) := \exp \left(\lambda y - \frac{\lambda^2}{2} \right),$$

so predictable choices of λ give the familiar exponential test supermartingales underlying time-uniform sub-Gaussian concentration (Howard et al., 2021). The one-step e-variable class is richer than this defining exponential family; recent work studies that geometry directly (Koolen et al., 2026).

At the level of the local model, sub-Gaussianity also lies beyond the finite-feature setting in (8). The local class $\mathcal{K}_{\text{SG}}$ is convex and compact and has a Borel graph. It is not locally dominated, and it is not representable by finitely many conditional expectations of a Borel feature map. Appendix E.4 applies Proposition E.2. The Gaussian law $\mathcal{N}(0, 1)$ makes every L_λ tight, while the functions

$$y \longmapsto e^{\lambda y - \lambda^2/2}, \quad \lambda \in \mathbb{R},$$

contain arbitrarily large linearly independent subfamilies. This gives the saturation obstruction formalized in Proposition E.2.

For these classical factors the robust one-step inequality is already exact under the Gaussian boundary law. Hence the corresponding exponential test martingales are fixed points of the Snell step in the supermartingale sense. Nonconstructivity enters when the theorem is applied to an arbitrary Borel e-process rather than to one of these explicit regular betting processes.

6.5 Example 4: a conditional Wasserstein ball

Let $(\mathcal{Y}, d)$ be an uncountable compact metric space, let $P_0 \in \Delta(\mathcal{Y})$ be a nominal law, write

$$D := \text{diam}(\mathcal{Y}),$$

and fix $\rho \in (0, D)$. Define the constant local correspondence

$$\mathcal{K}_W(s) := \{Q \in \Delta(\mathcal{Y}) : W_1(Q, P_0) \leq \rho\}.$$

The induced null requires

$$W_1(\mathcal{L}_P(Y_{t+1} \mid \mathcal{F}_t), P_0) \leq \rho$$

after every history. The conditional distribution may therefore change arbitrarily over time, provided each one-step law remains in the same transportation neighborhood of P_0 . This is a natural model of bounded distribution shift.

Because $W_1(\cdot, P_0)$ is continuous on $\Delta(\mathcal{Y})$ when $\mathcal{Y}$ is compact, $\mathcal{K}_W$ is compact and convex and its graph is Borel. It is locally nondominated for every P_0 and every $\rho > 0$ in the present uncountable compact setting; Appendix E.5 applies the common disjoint-support obstruction.

Whether a Wasserstein ball is beyond the finite-dimensional framework depends on its center. For example, if $P_0 = \delta_{y_0}$, then

$$W_1(Q, P_0) = \mathbb{E}_Q[d(Y, y_0)], \tag{9}$$

so the ball is a single expectation constraint. On the other hand, the usual continuous nominal models are genuinely richer. Appendix E.5 proves the following simple sufficient statement: on $\mathcal{Y} = [0, 1]$ with the Euclidean metric, if P_0 has a density bounded below by a positive constant on some nontrivial interval, then every sufficiently small positive-radius W_1 ball is not finitely feature-representable. The proof constructs one boundary law with infinitely many linearly independent tight Kantorovich–Rubinstein witnesses and then invokes the common saturation lemma.

The same duality gives a concrete test-supermartingale family. For every 1-Lipschitz f ,

$$\mathbb{E}_Q[f] - \mathbb{E}_{P_0}[f] \leq \rho \quad \text{for every } Q \in \mathcal{K}_W.$$

Hence

$$L_{f,\lambda}(y) := 1 + \lambda(f(y) - \mathbb{E}_{P_0}[f] - \rho), \quad 0 \leq \lambda \leq \frac{1}{D + \rho},$$

is a nonnegative one-step e-variable. Predictable choices of the Lipschitz witness and the stake therefore generate test supermartingales. An exponential version follows just as directly from Hoeffding’s lemma. These bets follow directly from Kantorovich–Rubinstein duality and are close in spirit to sequential betting based on learned witnesses for integral probability metrics (Shekhar and Ramdas, 2024). We have not found a previous reference that formulates a test supermartingale specifically for the conditional Wasserstein-ball null of Section 6.5; the construction itself is nevertheless a direct consequence of the same duality. Related work studies

Wasserstein uncertainty sets in robust fixed-sample testing and quickest change detection (Gao et al., 2018; Xie et al., 2024).

The Snell operator has an independent optimization interpretation here. For a continuation payoff $g_s(y) := g(s \oplus y)$,

$$(\mathbb{T}_t g)(s) = \sup_{Q: W_1(Q, P_0) \leq \rho} \mathbb{E}_Q[g_s].$$

For bounded continuous g_s , standard Wasserstein duality gives

$$(\mathbb{T}_t g)(s) = \inf_{\eta \geq 0} \left\{ \eta \rho + \int_{\mathcal{Y}} \sup_{y \in \mathcal{Y}} [g(s \oplus y) - \eta d(x, y)] P_0(dx) \right\};$$

see, for example, Esfahani and Kuhn (2018). Thus the dynamic primitive in the abstract Snell construction is the familiar worst-case expectation from Wasserstein distributionally robust optimization.

6.6 Example 5: latent feasibility and a genuinely analytic local model

Latent-variable and partial-identification models provide a natural statistical source of analytic admissibility conditions. A common structure is that an observed or reduced-form state is compatible with a model when there exists an unobserved parameter, latent variable, or structural object satisfying a measurable feasibility relation. Eliminating that latent witness is a projection operation. Even when the underlying relation is Borel, the resulting compatibility set need only be analytic. This issue appears explicitly in the partial-identification literature. For example, Kline and Tamer (2016, Remark 10) note that the set of reduced-form parameters for which an identified set is nonempty can fail to be Borel even when the criterion defining feasibility is measurable; their example is precisely a projection of a Borel set onto an analytic non-Borel set. Related random-set approaches to latent-variable identification rely on measurable selections from sets of latent values compatible with the observations, and universally measurable selections can arise naturally; see, for example, Beresteanu et al. (2012) and Gu and Russell (2023).

The following simple sequential model isolates this mechanism while keeping the one-step optimization explicit. Let $\mathcal{Y} = [0, 1]$, put

$$D_0 := [0, 1/2], \quad a := 1,$$

and let $\mathbf{Z}$ be a Polish space with a Borel relation $R \subseteq [0, 1] \times \mathbf{Z}$ whose projection

$$C := \{y : \text{there exists } z \in \mathbf{Z} \text{ with } (y, z) \in R\}$$

is analytic but non-Borel. One may read z as a latent feasibility certificate and C as the set of observed states compatible with at least one such certificate. The resulting admissibility rule is simple; the substantive feature is the projection mechanism rather than the binary form of the one-step set.

At the root let

$$\mathcal{K}_{\text{an}}(\emptyset) := \Delta(D_0),$$

and, for a nonempty history $s = (y_1, \dots, y_t)$, set

$$\mathcal{K}_{\text{an}}(s) := \begin{cases} \Delta(D_0 \cup \{a\}), & y_t \in C, \\ \Delta(D_0), & y_t \notin C. \end{cases}$$

Thus the additional outcome a is admissible precisely after histories whose most recent observation is compatible with some latent witness. Each value of the correspondence is compact and convex. Its graph, however, is analytic and genuinely non-Borel; Appendix E.6 gives the short verification and proves local nondomination using Proposition E.1.

This example cannot be represented by Clerico’s fixed finite-dimensional conditional model, for a measurability reason rather than the saturation argument of Proposition E.2. More generally, Proposition E.3 shows that a local model generated by finitely many jointly Borel history-dependent features together with a Borel feasible-set correspondence has a Borel graph. The genuinely non-Borel graph of $\mathcal{K}_{\text{an}}$ therefore rules out such a representation.

The effect on the Snell construction is visible from a single continuous continuation payoff. Let

$$\varphi(y) := (2y - 1)_+.$$

Then $\varphi = 0$ on D_0 and $\varphi(a) = 1$, so for every nonempty history $s = (y_1, \dots, y_t)$,

$$(\mathsf{T}_t \varphi)(s) = \sup_{Q \in \mathcal{K}_{\text{an}}(s)} \mathbb{E}_Q[\varphi(Y)] = \mathbf{1}_C(y_t).$$

The input is continuous, while the output is upper semianalytic and non-Borel. Moreover, even an ε -maximizing selector for this one-step problem, with $\varepsilon < 1/2$, need not admit a Borel version as a function of the history. A universally measurable selector does exist.

Thus the analytic machinery in the proof has a statistical interpretation as well as a technical one. Latent feasibility can turn a Borel relation on observed and unobserved objects into an analytic compatibility condition after the latent object is projected out. In the sequential setting this can make a continuous continuation payoff leave the Borel class under the upper transition operator and can force near-maximizing kernels into the universal completion. This sequential model abstracts that mechanism, while the underlying projection and measurable-selection issues have direct precedent in partial identification. The Borelization argument of Section 4 then shows that, despite these non-Borel intermediate objects, the final dominating test supermartingale can still be chosen Borel adapted.

Summary. Together, the five examples range from familiar finite-dimensional restrictions to nondominated and genuinely analytic conditional nulls. Bounded means connect the local-model language to standard betting and sit inside the finite-dimensional theory. FOSD and sub-Gaussianity are natural, nondominated, genuinely infinite-dimensional conditional nulls for which useful test supermartingales already exist. Wasserstein balls give a flexible distributional-stability model that is nondominated in full generality and beyond finite-dimensional constraints under a mild diffuse-center condition, while also admitting a direct Kantorovich–Rubinstein betting construction. The analytic example connects the same machinery to latent-feasibility and partial-identification problems, where projecting out an unobserved feasible object can produce analytic non-Borel compatibility sets; it then shows concretely why upper semianalytic dynamic programming and universal measurable selection can be necessary even for continuous continuation payoffs.

7 Discussion

We have shown that test supermartingales form a complete class for analytic sequential hypotheses: null families generated by measurable selections from an analytic correspondence of

admissible one-step laws. The result does not require local domination, convexity, or compactness of the local sets. Its proof separates two issues that are often hidden in regular examples. The robust Snell envelope supplies the right pointwise supermartingale inequalities, while reflection and monotone interlacing recover Borel measurability without increasing the initial testing budget.

The examples in Section 6 show that this scope includes both familiar and genuinely non-parametric conditional nulls. Bounded means recover the classical betting picture; FOSD and sub-Gaussianity lie beyond finite-feature conditional constraints; Wasserstein balls connect the upper transition operator to distributionally robust optimization; and latent feasibility gives a concrete reason for allowing analytic non-Borel local models.

These examples also separate structural completeness from the question of how explicitly the dominating process can be constructed. The main limitation of the general theorem is constructive. The measurable-selection and reflection steps are existential, so the dominating supermartingale supplied by the proof need not be directly computable from an arbitrary e-process. In regular examples the one-step upper expectation can be much more explicit—concavification for bounded means, or Wasserstein robust optimization for transport balls—and familiar test supermartingales may already be fixed points of the Snell operator. A natural next question is to identify broader conditions under which the complete-class result can be made operational: either through an explicit Snell operator, tractable approximate selectors, or a computable dominating supermartingale with controlled loss. Characterizing this constructive boundary is a natural direction for further work.

References

- Sebastian Arnold, Yo Joong Choe, Marco Scarsini, and Ilia Tsetlin. Betting on bets: Anytime-valid tests for stochastic dominance, 2026.
- Arie Beresteanu, Ilya Molchanov, and Francesca Molinari. Partial identification using random set theory. *Journal of Econometrics*, 166(1):17–32, 2012. doi: 10.1016/j.jeconom.2011.06.003.
- Dimitri Bertsekas and Steven E Shreve. *Stochastic optimal control: the discrete-time case*, volume 5. Athena Scientific, 1996.
- Bruno Bouchard and Marcel Nutz. Arbitrage and duality in nondominated discrete-time models. *Annals of Applied Probability*, 25(2):823–859, 2015. doi: 10.1214/14-AAP1011.
- Eugenio Clerico. On the optimality of coin-betting for mean estimation. *International Journal of Approximate Reasoning*, 187:109550, 2025. doi: 10.1016/j.ijar.2025.109550.
- Eugenio Clerico. Sequential testing of conditionally constrained hypotheses, 2026.
- Larry G. Epstein and Martin Schneider. Recursive multiple-priors. *Journal of Economic Theory*, 113(1):1–31, 2003. doi: 10.1016/S0022-0531(03)00097-8.
- Peyman Mohajerin Esfahani and Daniel Kuhn. Data-driven distributionally robust optimization using the wasserstein metric: Performance guarantees and tractable reformulations. *Mathematical Programming*, 171(1–2):115–166, 2018. doi: 10.1007/s10107-017-1172-1.
- Rui Gao, Liyan Xie, Yao Xie, and Huan Xu. Robust hypothesis testing using wasserstein uncertainty sets. In *Advances in Neural Information Processing Systems*, volume 31, pages 7913–7923, 2018.

- Peter Grünwald, Rianne de Heide, and Wouter M. Koolen. Safe testing. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 86(5):1091–1128, 2024. doi: 10.1093/jrsssb/qkae011.
- Jiaying Gu and Thomas M. Russell. Partial identification in nonseparable binary response models with endogenous regressors. *Journal of Econometrics*, 235(2):528–562, 2023. doi: 10.1016/j.jeconom.2022.04.011.
- Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, non-parametric, nonasymptotic confidence sequences. *Annals of Statistics*, 49(2):1055–1080, 2021. doi: 10.1214/20-AOS1991.
- Garud N. Iyengar. Robust dynamic programming. *Mathematics of Operations Research*, 30(2): 257–280, 2005. doi: 10.1287/moor.1040.0129.
- Olav Kallenberg. *Foundations of Modern Probability*. Springer, 3 edition, 2021. doi: 10.1007/978-3-030-61871-1.
- Alexander S. Kechris. *Classical Descriptive Set Theory*, volume 156 of *Graduate Texts in Mathematics*. Springer, 1995. doi: 10.1007/978-1-4612-4190-4.
- Brendan Kline and Elie Tamer. Bayesian inference in a class of partially identified models. *Quantitative Economics*, 7(2):329–366, 2016. doi: 10.3982/QE399.
- Wouter M. Koolen, Shubhada Agrawal, and Martin Larsson. A map of sub-gaussian e-variables. *ERCIM News*, (145), July 2026. URL <https://ercim-news.ercim.eu/en145/special/a-map-of-sub-gaussian-e-variables>.
- Martin Larsson, Aaditya Ramdas, and Johannes Ruf. Testing hypotheses generated by constraints. *Mathematics of Operations Research*, 2026. doi: 10.1287/moor.2025.1129. Published online May 12, 2026.
- Marcel Nutz and Ramon van Handel. Constructing sublinear expectations on path space. *Stochastic Processes and their Applications*, 123(8):3100–3121, 2013. doi: 10.1016/j.spa.2013.03.022.
- Goran Peskir and Albert Shiryaev. *Optimal Stopping and Free-Boundary Problems*. Birkhäuser, 2006. doi: 10.1007/978-3-7643-7390-0.
- Aaditya Ramdas, Johannes Ruf, Martin Larsson, and Wouter M. Koolen. Admissible anytime-valid sequential inference must rely on nonnegative martingales, 2020.
- Aaditya Ramdas, Johannes Ruf, Martin Larsson, and Wouter M. Koolen. Testing exchangeability: Fork-convexity, supermartingales and e-processes. *International Journal of Approximate Reasoning*, 141:83–109, 2022. doi: 10.1016/j.ijar.2021.06.017.
- Aaditya Ramdas, Peter Grünwald, Vladimir Vovk, and Glenn Shafer. Game-theoretic statistics and safe anytime-valid inference. *Statistical Science*, 38(4):576–601, 2023. doi: 10.1214/23-STS894.
- Glenn Shafer, Alexander Shen, Nikolai Vereshchagin, and Vladimir Vovk. Test martingales, bayes factors and p -values. *Statistical Science*, 26(1):84–101, 2011. doi: 10.1214/10-STS347.

- Shubhanshu Shekhar and Aaditya Ramdas. Nonparametric two-sample testing by betting. *IEEE Transactions on Information Theory*, 70(2):1178–1203, 2024. doi: 10.1109/TIT.2023.3305867.
- Albert N. Shiryaev. *Optimal Stopping Rules*. Springer, 2008. doi: 10.1007/978-3-540-74011-7. Reprint of the 1978 edition.
- Vladimir Vovk and Ruodu Wang. Merging sequential e-values via martingales. *Electronic Journal of Statistics*, 18(1):1185–1205, 2024. doi: 10.1214/24-EJS2228.
- Liyan Xie, Yuchen Liang, and Venugopal V. Veeravalli. Distributionally robust quickest change detection using wasserstein uncertainty sets. In *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*, pages 1063–1071. PMLR, 2024. URL <https://proceedings.mlr.press/v238/xie24a.html>.

A Path laws from universally measurable kernel selections

This appendix proves Lemma 2.2 and fixes the version conventions used in Section 3.

Proof of Lemma 2.2. Fix a $\mathcal{K}$ -selection $\kappa = (\kappa_t)_{t \geq 0}$. We recursively construct the finite-dimensional history laws. Set $\mu_0^\kappa := \delta_\emptyset$. Suppose μ_t^κ has been constructed on $\mathcal{Y}^t$. Since $\kappa_t : \mathcal{Y}^t \rightarrow \Delta(\mathcal{Y})$ is universally measurable and $\Delta(\mathcal{Y})$ is a standard Borel space, there exists a Borel measurable map $\bar{\kappa}_t : \mathcal{Y}^t \rightarrow \Delta(\mathcal{Y})$ such that $\bar{\kappa}_t = \kappa_t$ μ_t^κ -almost surely. Define

$$\mu_{t+1}^\kappa(B) := \int_{\mathcal{Y}^t} \int_{\mathcal{Y}} \mathbb{1}_B(s \oplus y) \bar{\kappa}_t(s, dy) \mu_t^\kappa(ds), \quad B \in \mathcal{B}(\mathcal{Y}^{t+1}).$$

Changing $\bar{\kappa}_t$ on a μ_t^κ -null set does not change μ_{t+1}^κ , so this recursion is independent of the chosen Borel version.

Applying the classical Ionescu–Tulcea theorem to the Borel kernels $(\bar{\kappa}_t)_{t \geq 0}$ gives a unique Borel probability measure P^κ on $(\Omega, \mathcal{F})$ whose t -dimensional marginal is μ_t^κ . Because $\bar{\kappa}_t = \kappa_t$ μ_t^κ -almost surely, the usual iterated-integral formula may equivalently be written using the original universally measurable kernels κ_t . Moreover, $\bar{\kappa}_t(Y_{1:t}, \cdot)$ is a regular conditional distribution of Y_{t+1} given $\mathcal{F}_t$ under P^κ , and $\kappa_t(Y_{1:t}, \cdot)$ is an almost-surely equal version on the P^κ -completion of $\mathcal{F}_t$. Uniqueness follows because the finite-dimensional cylinder sets generate $\mathcal{F}$. $\square$

B Measure-theoretic interface lemmas

B.1 Borel modification of universal stopping times

Lemma B.1 (Borel stopping-time modification). *Let P be a Borel probability measure on Ω , and let $\tau \in \mathcal{T}_b^u$. Then there exists a bounded stopping time σ with respect to the raw canonical filtration such that $P(\sigma = \tau) = 1$.*

Proof. Choose N such that $\tau \leq N$. For each $t \leq N$, write the exact-time event as $\{\tau = t\} = \{Y_{1:t} \in C_t\}$ for some $C_t \in \mathcal{U}_t$. Let P_t be the law of $Y_{1:t}$ under P . Because C_t belongs to the P_t -completion of $\mathcal{B}(\mathcal{Y}^t)$, choose a Borel set $D_t \subseteq \mathcal{Y}^t$ with $P_t(C_t \triangle D_t) = 0$. Set $\sigma := \inf\{0 \leq t \leq N : Y_{1:t} \in D_t\}$, with the convention that $\sigma = N$ if the set is empty. Then σ is a bounded Borel stopping time. Outside the P -null set $\bigcup_{t=0}^N \{Y_{1:t} \in C_t \triangle D_t\}$, membership in every exact-time region is unchanged, and hence $\sigma = \tau$. $\square$

B.2 Conditional expectations of universally measurable continuation values

Lemma B.2 (Conditional-kernel identity). *Fix a $\mathcal{K}$ -selection κ and $t \geq 0$. Let $f : \mathcal{Y}^t \times \mathcal{Y} \rightarrow [0, \infty]$ be universally measurable. Then*

$$\mathbb{E}_{P^\kappa} [f(Y_{1:t}, Y_{t+1}) \mid \mathcal{F}_t^{P^\kappa}] = \int_{\mathcal{Y}} f(Y_{1:t}, y) \kappa_t(Y_{1:t}, dy)$$

P^κ -almost surely, where $\mathcal{F}_t^{P^\kappa}$ denotes the P^κ -completion of $\mathcal{F}_t$.

Proof. We first treat bounded f . Let μ_t be the law of $Y_{1:t}$ under P^κ , and let $\bar{\kappa}_t$ be the Borel version of κ_t from Appendix A. The joint law π of $(Y_{1:t}, Y_{t+1})$ is $\pi(ds, dy) = \mu_t(ds) \bar{\kappa}_t(s, dy)$. Since f is universally measurable, it is measurable with respect to the π -completion of $\mathcal{B}(\mathcal{Y}^t \times \mathcal{Y})$, so there is a bounded Borel function $\bar{f}$ satisfying $f = \bar{f}$ π -almost surely. The ordinary Borel-kernel identity gives

$$\mathbb{E}_{P^\kappa} [\bar{f}(Y_{1:t}, Y_{t+1}) \mid \mathcal{F}_t^{P^\kappa}] = \int_{\mathcal{Y}} \bar{f}(Y_{1:t}, y) \bar{\kappa}_t(Y_{1:t}, dy)$$

P^κ -almost surely. The left-hand side is unchanged when $\bar{f}$ is replaced by f . On the right-hand side, Fubini–Tonelli applied to a Borel null superset of $\{f \neq \bar{f}\}$, together with $\kappa_t = \bar{\kappa}_t$ μ_t -almost surely, shows that

$$\int_{\mathcal{Y}} f(s, y) \kappa_t(s, dy) = \int_{\mathcal{Y}} \bar{f}(s, y) \bar{\kappa}_t(s, dy)$$

for μ_t -almost every s . This proves the bounded case. For general nonnegative f , apply the bounded result to $f \wedge m$ and use conditional monotone convergence. $\square$

B.3 Measurability of first contact

Lemma B.3 (First-contact stopping time). *Fix $N \in \mathbb{N}$ and let $E_t, M_t : \mathcal{Y}^t \rightarrow [0, \infty]$ be universally measurable for $t = 0, \dots, N$, with $M_N = E_N$. Then*

$$\tau^* := \inf\{0 \leq t \leq N : M_t(Y_{1:t}) = E_t(Y_{1:t})\}$$

belongs to $\mathcal{T}_b^u$.

Proof. For each $t \leq N$, the contact set $C_t := \{s \in \mathcal{Y}^t : M_t(s) = E_t(s)\}$ belongs to $\mathcal{U}_t$ because E_t and M_t are universally measurable. For every $j \leq N$, the event $\{\tau^* \leq j\}$ is the union, over $t \leq j$, of the pullbacks of C_t to $\mathcal{Y}^j$ under the corresponding prefix maps. Universal measurability is preserved under Borel preimages and finite unions, so $\{\tau^* \leq j\}$ is represented by a set in $\mathcal{U}_j$. Finally, $M_N = E_N$ ensures $\tau^* \leq N$. $\square$

C Analytic dynamic-programming facts

We collect the descriptive-set-theoretic facts used in Lemma 3.2. The following two standard facts are the basic inputs: if $h(x, y)$ is nonnegative **usa** on a product of Polish spaces, then $(x, Q) \mapsto \int h(x, y) Q(dy)$ is **usa**; and if $G \subseteq X \times A$ is analytic with nonempty sections and $h : X \times A \rightarrow [0, \infty]$ is **usa**, then $x \mapsto \sup\{h(x, a) : (x, a) \in G\}$ is **usa** and admits universally measurable δ -maximizing selectors when h is bounded. Both are standard consequences of the analytic measurable-selection machinery used in stochastic control; see Bertsekas and Shreve (1996, Chapter 7).

Proof of Lemma 3.2. Define $I(s, Q) := \int_{\mathcal{Y}} f(s \oplus y) Q(dy)$ on $\mathcal{Y}^t \times \Delta(\mathcal{Y})$. The integration fact stated at the start of Appendix C implies that I is **usa**. Since $\text{Gr}(\mathcal{K}) \cap (\mathcal{Y}^t \times \Delta(\mathcal{Y}))$ is analytic, for every $a \in \mathbb{R}$ the strict superlevel set of $\mathsf{T}_t f$ can be written as the projection

$$\{s : (\mathsf{T}_t f)(s) > a\} = \text{proj}_{\mathcal{Y}^t} (\text{Gr}(\mathcal{K}) \cap \{(s, Q) : I(s, Q) > a\}),$$

which is analytic. Thus $\mathsf{T}_t f$ is **usa**. Monotonicity is immediate from the definition. If $f_n \uparrow f$, monotone convergence gives, for each fixed s ,

$$\sup_n (\mathsf{T}_t f_n)(s) = \sup_n \sup_{Q \in \mathcal{K}(s)} \int f_n(s \oplus y) Q(dy) = \sup_{Q \in \mathcal{K}(s)} \int f(s \oplus y) Q(dy) = (\mathsf{T}_t f)(s),$$

where the two suprema commute because both sides are the supremum over $\mathbb{N} \times \mathcal{K}(s)$. Finally, when f is bounded, the measurable-selection fact stated at the start of Appendix C, applied to I on the analytic graph of $\mathcal{K}$, gives a universally measurable δ -maximizing selector κ_t^δ with the asserted inequality. $\square$

D Process reflection

This appendix proves Proposition 4.1. The argument uses a dual form of the First Reflection Theorem (Kechris, 1995, Theorem 35.10) together with a parameterized robust stopping problem.

Theorem D.1 (Dual First Reflection). *Let X be Polish and let Φ be a property of subsets of X which is Π_1^1 -on- Σ_1^1 : for every Polish parameter space Z and every analytic set $A \subseteq Z \times X$, the set $\{z \in Z : \Phi(A_z)\}$ is coanalytic. If an analytic set $D \subseteq X$ has property Φ , then there is a Borel set $B \supseteq D$ which also has property Φ .*

We next verify the parameterized measurability needed to apply this theorem to the robust stopped-value cost. For analytic $D \subseteq Z \times \mathsf{H} \times [0, \infty)$ and $z \in Z$, write $D_z := \{(s, u) : (z, s, u) \in D\}$. For $m \in \mathbb{N}$ and $s \in \mathcal{Y}^t$, define the truncated section length

$$g_t^m(z, s) := \int_0^m \mathbb{1}_D(z, s, u) du.$$

Lemma D.2 (Parameterized section lengths). *Let Z be Polish and let $D \subseteq Z \times \mathsf{H} \times [0, \infty)$ be analytic. For every $m \in \mathbb{N}$ and $t \geq 0$, the map $(z, s) \mapsto g_t^m(z, s)$ on $Z \times \mathcal{Y}^t$ is bounded and **usa**. Moreover, $g_{D_z}(s) = \sup_m g_t^m(z, s)$ for every $z \in Z$ and $s \in \mathcal{Y}^t$.*

Proof. The indicator of an analytic set is **usa**. Integrating a bounded **usa** function against the finite Borel measure $\lambda|_{[0, m]}$ preserves upper semianalyticity, so g_t^m is **usa** jointly in (z, s) . The final identity follows from monotone convergence of the section measures as $m \rightarrow \infty$. $\square$

Lemma D.3 (Parameterized robust stopped value). *Let Z be Polish and let $D \subseteq Z \times \mathsf{H} \times [0, \infty)$ be analytic. Then the map $z \mapsto \mathcal{C}(D_z)$ is **usa** on Z .*

Proof. Fix $m \in \mathbb{N}$ and a finite horizon N . Starting from the bounded **usa** reward functions g_t^m , define the parameterized Snell recursion by $V_N^{m, N}(z, s) := g_N^m(z, s)$ and, for $t < N$,

$$C_t^{m, N}(z, s) := \sup_{Q \in \mathcal{K}(s)} \int_{\mathcal{Y}} V_{t+1}^{m, N}(z, s \oplus y) Q(dy), \quad V_t^{m, N}(z, s) := g_t^m(z, s) \vee C_t^{m, N}(z, s).$$

The parameterized version of the analytic dynamic-programming facts in Appendix C shows inductively that $V_t^{m,N}$ and $C_t^{m,N}$ are **usa** jointly in (z, s) . In particular, $z \mapsto V_0^{m,N}(z, \emptyset)$ is **usa**.

For each fixed z , the usual finite-horizon robust stopping argument gives

$$V_0^{m,N}(z, \emptyset) = \sup_{\kappa} \sup_{\substack{\tau \in \mathcal{T}_b^u \\ \tau \leq N}} \mathbb{E}_{P^\kappa} [g_\tau^m(z, Y_{1:\tau})].$$

For completeness, the upper bound follows because the Snell envelope is a bounded supermartingale under every P^κ and dominates the reward. For the reverse bound, choose universally measurable δ_t -maximizing kernels and stop in the universally measurable regions

$$\{s : g_t^m(z, s) \geq C_t^{m,N}(z, s) - \delta_t\},$$

with $\sum_{t < N} \delta_t$ arbitrarily small; the standard backward ε -optimality induction then gives the reverse inequality. Here z is fixed before the policy is selected, so no selector jointly measurable in (z, s) is required.

Every bounded stopping time is bounded by some deterministic horizon, and for fixed κ and τ , monotone convergence gives

$$\mathbb{E}_{P^\kappa} [g_{D_z}(Y_{1:\tau})] = \sup_m \mathbb{E}_{P^\kappa} [g_\tau^m(z, Y_{1:\tau})].$$

Consequently,

$$\mathcal{C}(D_z) = \sup_{N \geq 0} \sup_{m \geq 1} V_0^{m,N}(z, \emptyset).$$

This is a countable supremum of **usa** functions of z , and is therefore **usa**. $\square$

Proof of Proposition 4.1. Let $q := \mathcal{C}(D) < \infty$. For analytic subsets $S \subseteq \mathbf{H} \times [0, \infty)$, consider the property $\Phi_q(S)$ defined by $\mathcal{C}(S) \leq q$. To verify that this property is Π_1^1 -on- Σ_1^1 , let Z be Polish and $A \subseteq Z \times \mathbf{H} \times [0, \infty)$ be analytic. By Lemma D.3, the map $z \mapsto \mathcal{C}(A_z)$ is **usa**, so

$$\{z : \Phi_q(A_z)\} = \{z : \mathcal{C}(A_z) \leq q\}$$

is coanalytic. Theorem D.1 therefore yields a Borel set $B \supseteq D$ with $\mathcal{C}(B) \leq q$. Monotonicity of $\mathcal{C}$ under inclusion gives $q = \mathcal{C}(D) \leq \mathcal{C}(B)$, hence $\mathcal{C}(B) = \mathcal{C}(D)$. $\square$

E Technical details for the examples

The structural lemmas and technical verifications for Section 6 are collected here. Propositions E.1, E.2, and E.3 state the reusable obstructions, followed by the examples in the same order as in Section 6.

E.1 Reusable structural lemmas

Two structural criteria recur across the examples. The first rules out local domination. The second rules out finite-dimensional conditional-constraint representations.

Proposition E.1 (A nondomination criterion). *Let $\mathcal{K}_0 \subseteq \Delta(\mathcal{Y})$. Suppose there is an uncountable index set A , laws $\{Q_\alpha : \alpha \in A\} \subseteq \mathcal{K}_0$, and pairwise disjoint measurable sets $\{B_\alpha : \alpha \in A\}$ such that*

$$Q_\alpha(B_\alpha) > 0 \quad \text{for every } \alpha \in A.$$

Then no σ -finite measure dominates every law in $\mathcal{K}_0$.

Proof. Suppose a σ -finite measure ν dominated every Q_α . Then $Q_\alpha(B_\alpha) > 0$ implies $\nu(B_\alpha) > 0$ for every α . Write $\mathcal{Y} = \bigcup_{n \geq 1} A_n$ with $\nu(A_n) < \infty$. For every α , some n satisfies $\nu(B_\alpha \cap A_n) > 0$. For fixed n , the pairwise disjoint sets $B_\alpha \cap A_n$ can have positive ν -mass for at most countably many α . Taking the countable union over n shows that only countably many B_α can have positive mass, contradicting uncountability of A . $\square$

The special case used in Appendices E.2, E.4, E.5, and E.6 takes B_α to be distinct singletons.

Proposition E.2 (Saturating e-variables obstruct finite-dimensional representations). *Let $\mathcal{K}_0 \subseteq \Delta(\mathcal{Y})$ be a one-step null on a standard Borel space. Suppose there is a law $Q_0 \in \mathcal{K}_0$ and a family of nonnegative Borel functions $\{E_\alpha : \alpha \in A\}$ such that*

$$\sup_{Q \in \mathcal{K}_0} \mathbb{E}_Q[E_\alpha] \leq 1, \quad \mathbb{E}_{Q_0}[E_\alpha] = 1$$

for every α , and such that the span of $\{E_\alpha : \alpha \in A\}$ is infinite-dimensional modulo Q_0 -almost-sure equality. Then there do not exist a finite d , a Borel map $\Phi : \mathcal{Y} \rightarrow \mathbb{R}^d$, and a nonempty convex set $S \subseteq \mathbb{R}^d$ for which

$$\mathcal{K}_0 = \{Q : \mathbb{E}_Q[\|\Phi\|_1] < \infty, \quad \mathbb{E}_Q[\Phi] \in S\}.$$

Proof. Assume such a representation exists. For each α , the one-step affine-domination result of Clerico (2026) gives a valid e-variable $\hat{E}_\alpha$ dominating E_α pointwise on the effective domain and having the form

$$\hat{E}_\alpha = a_\alpha + b_\alpha^\top \Phi.$$

Since Q_0 belongs to the null,

$$1 = \mathbb{E}_{Q_0}[E_\alpha] \leq \mathbb{E}_{Q_0}[\hat{E}_\alpha] \leq 1.$$

Hence $\hat{E}_\alpha - E_\alpha \geq 0$ has zero Q_0 -expectation, so

$$E_\alpha = \hat{E}_\alpha \quad Q_0\text{-almost surely.}$$

Thus every E_α belongs, modulo Q_0 -null sets, to

$$\text{span}\{1, \Phi_1, \dots, \Phi_d\},$$

a vector space of dimension at most $d + 1$. This contradicts the assumed infinite-dimensionality of the witness family. $\square$

Proposition E.2 is the common argument used for FOSD (Appendix E.3), sub-Gaussianity (Appendix E.4), and the nondegenerate Wasserstein balls (Appendix E.5). The analytic example uses a different obstruction.

Proposition E.3 (Borel finite-feature models have Borel graphs). *Let $\Phi : \mathbf{H} \times \mathcal{Y} \rightarrow \mathbb{R}^d$ be Borel, and let $S : \mathbf{H} \rightrightarrows \mathbb{R}^d$ have Borel graph. Define*

$$\mathcal{K}_{\Phi, S}(s) := \left\{ Q \in \Delta(\mathcal{Y}) : \int \|\Phi(s, y)\|_1 Q(dy) < \infty, \quad \int \Phi(s, y) Q(dy) \in S(s) \right\}.$$

Then $\text{Gr}(\mathcal{K}_{\Phi, S})$ is Borel in $\mathbf{H} \times \Delta(\mathcal{Y})$. In particular, a local correspondence with genuinely non-Borel graph cannot admit such a representation.

Proof. For every nonnegative Borel function $h(s, y)$, the map

$$(s, Q) \mapsto \int h(s, y) Q(dy)$$

is Borel as an extended-real-valued function; this follows first for indicators of Borel rectangles, then for bounded Borel functions by a monotone-class argument, and finally for nonnegative functions by monotone convergence. Applying this to the positive and negative parts of the coordinates of Φ , and to $\|\Phi\|_1$, shows that the integrability set is Borel and that the vector integral is Borel on that set. Membership of the integral in $S(s)$ is then the inverse image of $\text{Gr}(S)$ under a Borel map. Hence $\text{Gr}(\mathcal{K}_{\Phi, S})$ is Borel. $\square$

E.2 Bounded conditional means

The graph and compactness claims in Example 6.1 follow from the continuity of

$$Q \mapsto \int y Q(dy)$$

under weak convergence on $\Delta([0, 1])$. Since $[0, 1]$ is compact, $\Delta([0, 1])$ is weakly compact, and the level set at μ is therefore compact.

For nondomination, for each $x \in [0, \mu)$ define

$$p_x := \frac{1 - \mu}{1 - x}, \quad Q_x := p_x \delta_x + (1 - p_x) \delta_1.$$

Then $Q_x \in \mathcal{K}_\mu$ and $Q_x(\{x\}) = p_x > 0$. The singletons $\{x\}$ are pairwise disjoint for uncountably many x , so Proposition E.1 applies.

For completeness, the concave-envelope identity used in Section 3.1 can be written as

$$\sup_{Q: \mathbb{E}_Q Y = \mu} \mathbb{E}_Q[f(Y)] = \text{conc}(f)(\mu)$$

for bounded continuous $f : [0, 1] \rightarrow \mathbb{R}$. The upper bound follows from Jensen's inequality applied to every concave majorant. For the reverse inequality, the least concave majorant at μ is the supremum of chord values

$$\frac{b - \mu}{b - a} f(a) + \frac{\mu - a}{b - a} f(b), \quad 0 \leq a \leq \mu \leq b \leq 1,$$

with the obvious interpretation when $a = b = \mu$. The corresponding two-point distribution has mean μ .

E.3 Conditional first-order stochastic dominance

For $Q \in \Delta([0, 1]^2)$,

$$Q_Z \preceq_{\text{FOSD}} Q_X$$

is equivalent to

$$\int \phi dQ_X \geq \int \phi dQ_Z$$

for every bounded continuous increasing $\phi : [0, 1] \rightarrow \mathbb{R}$. Each inequality defines a weakly closed half-space, so $\mathcal{K}_{\text{FOSD}}$ is closed and convex. It is compact because $[0, 1]^2$ is compact. Hence the constant correspondence has Borel graph.

For nondomination, every point mass $\delta_{(x,z)}$ with $x \geq z$ belongs to the null. The uncountable family of singletons in the triangle

$$\{(x, z) \in [0, 1]^2 : x \geq z\}$$

therefore satisfies Proposition E.1.

We next verify the hypotheses of Proposition E.2. Fix $0 < \eta \leq 1$ and, for $c \in (0, 1)$, define

$$E_c(x, z) := 1 + \eta (\mathbb{1}\{x \leq c\} - \mathbb{1}\{z \leq c\}).$$

The factor is nonnegative, and for every $Q \in \mathcal{K}_{\text{FOSD}}$,

$$\mathbb{E}_Q[E_c] = 1 + \eta (F_{Q_X}(c) - F_{Q_Z}(c)) \leq 1.$$

Let

$$Q_0 := \text{Unif}[0, 1] \otimes \text{Unif}[0, 1].$$

Its marginals agree, so Q_0 is in the null and $\mathbb{E}_{Q_0}[E_c] = 1$ for every c .

It remains to show infinite-dimensionality modulo Q_0 . Take distinct $c_1, \dots, c_n$ and suppose

$$\sum_{j=1}^n a_j E_{c_j}(x, z) = 0$$

for Lebesgue-almost every (x, z) . Set

$$g(u) := \sum_{j=1}^n a_j \mathbb{1}\{u \leq c_j\}.$$

Then

$$\sum_{j=1}^n a_j + \eta (g(x) - g(z)) = 0$$

for almost every (x, z) . Interchanging x and z and adding gives $\sum_j a_j = 0$, after which $g(x) = g(z)$ for almost every pair. Thus g is almost everywhere constant. It vanishes above $\max_j c_j$, so it is zero almost everywhere. The jump at each distinct c_j has size $-a_j$, hence every $a_j = 0$. Proposition E.2 now shows that $\mathcal{K}_{\text{FOSD}}$ has no finite-dimensional representation of Clerico's form.

E.4 Conditional sub-Gaussianity

For fixed $\lambda \in \mathbb{R}$, the map

$$Q \mapsto \int e^{\lambda y} Q(dy)$$

is lower semicontinuous under weak convergence. Hence the sublevel set at $e^{\lambda^2/2}$ is closed, and arbitrary intersection over λ shows that $\mathcal{K}_{\text{SG}}$ is closed. It is convex by linearity of expectation. Moreover, the MGF bound gives the uniform tail estimate

$$Q(|Y| \geq r) \leq 2e^{-r^2/2}, \quad Q \in \mathcal{K}_{\text{SG}},$$

so the class is uniformly tight. Prokhorov's theorem therefore gives compactness, and the constant local correspondence has Borel graph.

For nondomination, for every $x \in (0, 1]$ let

$$Q_x := \frac{1}{2}\delta_{-x} + \frac{1}{2}\delta_x.$$

Then

$$\mathbb{E}_{Q_x}[e^{\lambda Y}] = \cosh(\lambda x) \leq e^{\lambda^2 x^2/2} \leq e^{\lambda^2/2},$$

so $Q_x \in \mathcal{K}_{\text{SG}}$. The singletons $\{x\}$ give the disjoint sets required by Proposition E.1.

For the finite-dimensional obstruction, define

$$E_\lambda(y) := \exp\left(\lambda y - \frac{\lambda^2}{2}\right).$$

These are e-variables for $\mathcal{K}_{\text{SG}}$ by definition. Let $\gamma = N(0, 1)$. Then $\gamma \in \mathcal{K}_{\text{SG}}$ and

$$\mathbb{E}_\gamma[E_\lambda] = 1$$

for every λ . To see linear independence, take distinct $\lambda_1, \dots, \lambda_n$ and suppose

$$\sum_{j=1}^n a_j e^{\lambda_j y - \lambda_j^2/2} = 0$$

for γ -almost every y . The left side is continuous and γ has a strictly positive density, so the identity holds for every y . Absorb the constants $e^{-\lambda_j^2/2}$ into the coefficients and differentiate $k = 0, \dots, n-1$ times at $y = 0$. The resulting Vandermonde system is nonsingular because the λ_j are distinct. Thus all coefficients vanish. Proposition E.2 applies.

E.5 Conditional Wasserstein balls

Regularity and nondomination. On a compact metric space, W_1 metrizes weak convergence. Hence

$$\mathcal{K}_W = \{Q : W_1(Q, P_0) \leq \rho\}$$

is weakly closed and therefore compact; it is convex by convexity of $W_1(\cdot, P_0)$. The graph of the constant correspondence is Borel.

For nondomination, fix

$$0 < \varepsilon < \frac{\rho}{D}$$

and, for $x \in \mathcal{Y}$, set

$$Q_x := (1 - \varepsilon)P_0 + \varepsilon\delta_x.$$

Coupling the common $(1 - \varepsilon)P_0$ part identically and moving only the remaining mass gives

$$W_1(Q_x, P_0) \leq \varepsilon D < \rho.$$

A probability measure has at most countably many atoms, so there is an uncountable set $A \subseteq \mathcal{Y}$ on which $P_0(\{x\}) = 0$. For every $x \in A$,

$$Q_x(\{x\}) = \varepsilon > 0.$$

Proposition E.1 applies to the singletons $\{x\}$.

A clean sufficient condition for being beyond finite-dimensional constraints. We give a concrete statement in the standard one-dimensional setting. Let $\mathcal{Y} = [0, 1]$ with $d(x, y) = |x - y|$. Suppose there are an interval $I = [a, b]$ of length $L = b - a > 0$ and a constant $c > 0$ such that

$$P_0(A) \geq c \text{Leb}(A) \quad \text{for every Borel } A \subseteq I.$$

In particular, this holds if P_0 has a density bounded below by a positive constant on a nontrivial interval. We show that if

$$0 < \rho < \frac{cL^2}{16},$$

then $\mathcal{K}_W$ is not finitely feature-representable.

Set $\ell = L/4$ and define adjacent intervals

$$I_1 = [a, a + \ell], \quad I_2 = [a + \ell, a + 2\ell].$$

Let

$$\alpha := \frac{\rho}{\ell^2} < c,$$

and define the probability measure

$$Q_\star := P_0 - \alpha \text{Leb}|_{I_1} + \alpha \text{Leb}|_{I_2}.$$

The lower bound on P_0 ensures nonnegativity on I_1 . Using the one-dimensional formula

$$W_1(P, Q) = \int_0^1 |F_P(u) - F_Q(u)| du,$$

the CDF difference generated by this transfer is a triangular excursion over $I_1 \cup I_2$ with area $\alpha\ell^2$. Hence

$$W_1(Q_\star, P_0) = \rho.$$

Choose a 1-Lipschitz function f_0 that has slope 1 on $I_1 \cup I_2$ and is constant on $[a + 2\ell, b]$. For example, after adding an irrelevant constant, one may take

$$f_0(y) = \min\{\max\{y - a, 0\}, 2\ell\}.$$

A direct calculation gives

$$\mathbb{E}_{Q_\star}[f_0] - \mathbb{E}_{P_0}[f_0] = \rho,$$

so f_0 is a Kantorovich–Rubinstein witness for the boundary pair.

There is still a nontrivial interval

$$J := (a + 2\ell, b)$$

on which $Q_\star = P_0$ and f_0 is constant. Choose pairwise disjoint nonempty open intervals $J_k \subset J$ and nonzero tent functions h_k supported on J_k with Lipschitz constant at most $1/2$. After scaling the tents if necessary, the functions

$$f_k := f_0 + h_k$$

are 1-Lipschitz. Because $Q_\star - P_0$ is supported on $I_1 \cup I_2$,

$$\mathbb{E}_{Q_\star}[f_k] - \mathbb{E}_{P_0}[f_k] = \rho$$

for every k .

Fix a sufficiently small common $\lambda > 0$ and define

$$E_k(y) := 1 + \lambda(f_k(y) - \mathbb{E}_{P_0}[f_k] - \rho).$$

Since $[0, 1]$ is compact and the f_k are uniformly bounded after the common normalization, λ can be chosen so that all E_k are nonnegative. Kantorovich–Rubinstein duality gives, for every $Q \in \mathcal{K}_W$,

$$\mathbb{E}_Q[E_k] \leq 1,$$

while at the boundary law

$$\mathbb{E}_{Q_\star}[E_k] = 1.$$

Finally, the family $\{E_k\}$ is linearly independent modulo $Q_\star$. Indeed, on J we have $Q_\star = P_0 \geq c$ Leb and the supports of the h_k are disjoint. For any finite relation $\sum_{j=1}^m a_j E_{k_j} = 0$ $Q_\star$ -almost surely, first evaluate on a positive-measure subset of J outside all of the bump supports to obtain the common baseline relation. Then restrict to J_{k_j} and subtract that baseline relation; this leaves

$$a_j \lambda h_{k_j} = 0$$

almost everywhere on J_{k_j} , so $a_j = 0$. Proposition E.2 completes the proof.

The diffuse-center condition used in Appendix E.5 is only sufficient, not necessary. It is included to make precise a broad natural regime in which the Wasserstein ball is beyond the finite-dimensional framework. By (9), a ball centered at a point mass can instead reduce to a single expectation constraint.

E.6 Latent feasibility and the analytic example

For $t \geq 1$, let

$$C_t := \{s = (y_1, \dots, y_t) : y_t \in C\}.$$

The last-coordinate map is continuous, so C_t is analytic and non-Borel whenever C is. For a closed set $F \subseteq [0, 1]$, define

$$\Delta(F) := \{Q \in \Delta([0, 1]) : Q(F) = 1\}.$$

This is weakly closed. The graph at time t can be written as

$$\text{Gr}_t(\mathcal{K}_{\text{an}}) = (\mathcal{Y}^t \times \Delta(D_0)) \cup (C_t \times \Delta(D_0 \cup \{a\})).$$

The first term is Borel and the second is analytic, so the graph is analytic. It is not Borel, because its section at the fixed measure δ_a is exactly C_t :

$$\{s : (s, \delta_a) \in \text{Gr}_t(\mathcal{K}_{\text{an}})\} = C_t.$$

Local nondomination follows immediately because every $\mathcal{K}_{\text{an}}(s)$ contains $\Delta(D_0)$, hence contains δ_x for every $x \in D_0$. Proposition E.1 applies. Proposition E.3 also shows that this genuinely non-Borel correspondence cannot be generated by a finite-dimensional jointly Borel feature map and a Borel feasible-set correspondence. In particular it is outside the fixed finite-dimensional conditional model of Clerico (2026).

Finally, for the continuous payoff $\varphi(y) = (2y - 1)_+$,

$$(\mathsf{T}_t \varphi)(s) = \mathbb{1}_{C_t}(s).$$

Thus the upper transition is non-Borel. To see that approximate maximizers may also require universal measurability, fix $\varepsilon \in (0, 1/2)$. Any ε -optimal selector κ_t must satisfy

$$\int \varphi d\kappa_t(s) > \frac{1}{2} \iff s \in C_t.$$

The map $Q \mapsto \int \varphi dQ$ is continuous, so

$$U := \left\{ Q : \int \varphi dQ > \frac{1}{2} \right\}$$

is open in $\Delta(\mathcal{Y})$. If κ_t were Borel, then $C_t = \kappa_t^{-1}(U)$ would be Borel, a contradiction. A universally measurable selector exists because analytic sets are universally measurable, and more generally by the Jankov–von Neumann theorem.